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Feature selection techniques in the context of big data: taxonomy and analysis

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Abstract

Recent advancements in Information Technology (IT) have engendered the rapid production of big data, as enormous volumes of data with high dimensional features grow exponentially in different fields. Therefore, dealing with high-dimensional data creates new challenges in terms of data processing efficiency and effectiveness. To address such challenges, Feature Selection (FS) is among the most utilized dimensionality reduction methods, which is helpful in reducing the high dimensionality of large-scale data by picking up a small subset of related and significant features and eliminating unrelated and redundant features in order to construct effective prediction models. This article provides a comprehensive review of the latest FS approaches in the context of big data along with a structured taxonomy, which categorizes the existing methods based on their nature, search strategy, evaluation process, and feature structure. Moreover, it presents a qualitative analysis of FS methods based on their objective, structure, search strategy, schema, learning task, strengths, and weaknesses. Further, a quantitative analysis is also performed to illustrate the number of publications related to FS based on the timeline, main category, and other sub-categories. An experimental study is also conducted comparing ten methods from different categories using twelve benchmark datasets from the University of California, Irvine (UCI) Machine Learning Repository and Arizona State University (ASU) Feature Selection Repository to evaluate their performance in terms of (accuracy, precision, recall, F-measures, and the number of selected features). Finally, we highlight the research issues and open challenges related to FS to assist researchers in identifying future research directions.

Keywords Big Data · Dimensionality Reduction · Feature Selection · Streaming Feature

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1 Introduction

The advancement in the Information Technology (IT) age has ushered in a new era of data analysis, where a huge amount of data is extremely generated daily with high dimensionality in various fields such as business intelligence, healthcare, social media, transportation, online education, government, marketing, financial, etc. Such data with high complexity, huge volume, and fast-growing rate from multiple, autonomous sources is termed as "Big Data" [1]. These large-scale data contain a lot of insights and knowledge, which is of great value and importance to decision-makers. Extracting such insights and knowledge is a challenging task in machine learning and data mining [2, 3]; therefore, there is a need for efficient and effective data analytics techniques. The existing data analytics techniques are not ideal for extracting the latent data insights and knowledge due to their inability of handling large-scale data with high dimensionality [4]. Big data with large dimensions leads to redundant and unrepresentative features that greatly increase memory storage needs, add additional processing time to analytics techniques, and thus negatively affect their performance and increase data processing expenses.

Dimensionality reduction is the key task and powerful technique for data pre-processing [5], which attempts to map space of high dimensionality into lower-dimensionality without substantial loss of information. Dimensionality reduction methods include two types: Feature Extraction and FS. The feature extraction methods tend to map the original features into lower dimensions with a new feature space; the newly added features generally are a combination of original features. On the other hand, FS tends to pick up a significant small subset of features from the original dataset by removing irrelevant, redundant, or noisy features based on a predefined evaluation measure [6]. Both methods aim at reducing the dimensionality in a large dataset, but the feature extraction methods do this by generating new feature combinations, whereas, FS chooses a feature subset from all the available features without any modifications. Therefore, the FS methods are often recommended for several domains and applications such as text mining and genetic analysis, where the readability and interpretability of the data are necessary because the meaning of the original features is kept in the reduced subset.

FS is among the most important pre-processing methods in data mining, machine learning, and pattern recognition that plays an essential role in extracting valuable information from a large-scale dataset. The feature selection primary task is to identify a significant feature subset from the large data set and eliminate those irrelevant and redundant features without any significant loss of information or degradation of learning performance, which results in better prediction accuracy [7]. FS offers many benefits such as minimizing the storage needs, improving learning performance, higher prediction accuracy, and minimal computational costs [8]. For this reason, FS has been an active research area in data analytics and pattern recognition where many methods have been proposed in the literature, but still they should be more efficient, capable of handling high dimensional data [9].

Many surveys have been presented in the literature to review the existing FS methods. Such as the survey proposed by Chandrashekar et al. [10] which presented an overview of the FS and categorized them into filter, wrapper, and embedded methods. Bolón-Canedo et al. [8] provided a comprehensive study of different FS methods for classification problems and also discussed the basics, applications, and the challenges in the context of high-dimensional data. Sheikhpour et al. [11] presented a comprehensive study focused on semi-supervised FS methods. Cai et al. [12] proposed an overview of several commonly used evaluation measures and then surveyed some supervised, unsupervised, and semi-supervised FS approaches. Venkatesh et al. [13] presented a survey on FS methods that uses static data and their potential applications. Solorio-Fernández et al. [14] introduced a comprehensive review of recent advanced unsupervised FS methods. Deng et al. [15] presented a brief review of FS techniques used in text classification. Remeseiro et al. [16] proposed an overview of FS approaches used in medical applications.

Some other few surveys discussed the methods of Online Feature Selection (OFS) such as [17–19]. Wang et al. [17] proposed a survey on FS techniques for classification problems. Li et al. [18] presented a comprehensive review of recent advanced FS methods from a data perspective view. Rong et al. [19] presented a comprehensive survey of FS techniques used in big data. These surveys still lack in-depth discussions and are not up to date, they neglected the Online Group Feature Selection (OGFS). Hu et al. [20] provided a comprehensive review of existing OFS with streaming features methods along with their merits and demerits. This survey focused on OFS with streaming features, they have not considered OFS with streaming data, since this paper was published in the year 2015, therefore, still there is a lack of in-depth discussions on the most recent methods. AlNuaimi et al. [21] presented a survey on FS methods, they have classified the methods into static and streaming FS methods, but they neglected to review some of the recent Swarm Intelligence (SI) based FS algorithms. Moreover, this survey lacks Experimental analysis. Solorio-Fernández et al. [22] proposed a comprehensive and systematic review of the most popular and recent filter unsupervised FS methods, evaluating their performance. This survey has particularly focused on filter FS methods only. Whereas the Sharma et al. [23] specifically, focus on nature-inspired meta-heuristic methods and their role in the FS process.

Sanagavarapu et al. [24] presented an analysis for three FS methods evaluating their accuracy using in order to find out the efficient method. Solorio-Fernández et al. [25] presented a comparative study to evaluate seven of the most commonly used and outstanding ranking-based unsupervised FS methods, which are under the category of filters. They have evaluated the methods based on the quality of selection and runtime. Bommert et al. [26] performed a comparative study on 22 state-of-the-art filter-based FS methods. This article analyzes the FS filter methods based on runtime and prediction accuracy. Rostami et al. [27] reviewed SI-based FS methods and provided a comparative analysis based on their performance.

Though there are many exisitng surveys, but they are limited to Traditional FS techniques or specific domains applications such as medical application and text classification, and some of them did not cover the topic of OFS. There is no single survey comprehensively covered and analyzed all categories of FS, qualitatively, quantitatively and experimentally. Therefore, this is the major motivagion behind presenting this comprehensive taxonomy survey and empirical analysis, which provide a broader and integrative classification of the most recent, ever-growing effective FS methods.

This study aims at classifiing the FS techniques broadly into two main categories Traditional Feature Selection (TFS) and Online Feature Selection (OFS). Moreover, it presents a qualitative analysis of FS technqiues based on their structure, objective, search strategy, schema, learning task, strengths, and weaknesses. Further, a quantitative analysis is also performed to illustrate the number of publications related to FS based on the timeline, main category, and other sub-categories. An experimental study is also conducted comparing ten methods from different categories using twelve benchmark datasets from the UCI Machine Learning Repository and AUS Feature Selection Repository to evaluate their performance in terms of (accuracy, precision, recall, F-measures, and the number of selected features). This article also highiligtes research issues and open challenges related to FS to help researchers to understand the basics concepts of FS and get insight about the existing literature of FS. It also shapes the way for future research directions and serves as an opportunity for researchers to focus on new ideas for research.

1.1 Paper contributions

The main contributions of the taxonomy survey are outlined as following:

- A review and comparative analysis of the existing surveys on FS, highlighting their contributions and limitations.
- A comprehensive survey along with structured taxonomy that classifies FS into various categories based on their nature, search strategy, and evaluation criteria.
- A qualitative analysis of the reviewed FS methods based on their objective, schema, search strategy, learning task strength, and weaknesses.
- A quantitative analysis of the reviewed FS methods based on their publication timeline, main category, and other sub-categories.
- Experimental analysis on some of the prominent filter and wrapper methods with three classifiers with 12 datasets.
- A highlight of some current research challenges and open issues as previously indicated in prior research studies, which helps to influence the direction of future research.

This taxonomy survey boosts the existing surveys and taxonomies, it also proposes a comprehensive review to bring a light on the research challenges in FS, which can be helpful for researchers to understand and apply the vital characteristics of FS in Big Data. It also aids them in finding out the limitations of current FS methods and choose a significantly possible future research direction.

1.2 Paper organization

The paper is organized as the following: Section 2 discusses the existing surveys, then analyzes their contributions and limitations. Section 3 provides a taxonomy of the most recent existing FS methods and reviews them based on the taxonomy. Section 4 presents a quantitative analysis of the exiting FS methods. Section 5 performed an experimental analysis of some of the most prominent filter and wrapper methods. Section 6 highlights the research challenges and open issues. Section 7 concludes the article and brings light on the future work.

2 Existing surveys

In this section, we review and analyze the current existing surveys related to our taxonomy. Generally, several surveys and reviews are being proposed for FS in the literature. Some of these surveys focus either on TFS techniques or specific learning tasks as supervised, unsupervised and semi-supervised [8, 10–14], whereas other studies focus on describing the FS methods applied to particular domains and applications [15, 16], however, none of the above mentioned existing surveys have highlighted the OFS algorithms, as it is clearly shown the Table 2. Only a few surveys discussed the methods of OFS especially in the context of online individual feature selection such as [17–19], but they still lack in-depth discussions and are not up to date, they neglected the online group feature selection. In [20] the OFS with streaming features were particularly considered, they have not considered OFS with streaming data, this survey paper was published in the year 2015, therefore, still there is a lack of in-depth discussions on the most recent methods. Authors in [21] presented a survey on FS methods, they have classified the methods into static and Streaming FS methods, but they neglected to review some of the recent search-based methods of Swarm Intelligence (SI) such as Particle Swarm Optimization (PSO), Whale Optimization Algorithm (WOA), Cuckoo Search (CS), Bat Algorithm (BA), Firefly Algorithm (FA), Ant Colony Optimization (ACO), moreover, this survey lacks Experimental analysis. Authors in [22] proposed a comprehensive and systematic review of the most popular and recent filter unsupervised FS methods, evaluating their performance based on classification, clustering, and runtime. This survey has particularly focused on filter FS methods only. Whereas the Authors in [23] specifically,

Table 1 summary of the existing surveys based on contribution.

Year	Ref.	Contribution
2014	[10]	Presented a survey on FS for machine learning problems, they focus on Filter, Wrapper, and Embedded Methods.
2014	[17]	Proposed a survey on FS techniques for classification problems.
2016	[8]	Provided a comprehensive study of different FS methods for classification problems, they also discussed the basics, real applications, and the challenges in the context of high-dimensional data.
2017	[11]	Provided a comprehensive study on semi-supervised FS methods, classifies the techniques from two different perspectives.
2018	[18]	Presented a comprehensive review of recent advanced FS methods from a data perspective view.
2018	[12]	Presented overview of several commonly used evaluation measures and then surveyed some supervised, unsupervised, and semi-supervised FS approaches.
2018	[20]	Provided a comprehensive review of existing OFS methods along with their merits and demerits.
2018	[15]	Provided a brief review of FS techniques used in text classification.
2019	[19]	Presented a comprehensive survey of FS techniques used in big data.
2019	[13]	Provided a survey on FS methods that uses static data and their potential applications.
2019	[21]	Provided a survey on different FS methods and reviewed the current methods that use OFS highlighting their strengths and weaknesses.
2019	[16]	Provided an overview of FS approaches used for medical applications.
2020	[14]	Presented a comprehensive review of recent advanced unsupervised FS methods.
2020	[22]	Presented a comprehensive and systematic review of the most popular and recent filter unsupervised FS methods.
2020	[23]	Presented a survey on nature-inspired meta-heuristic methods and their role in the FS process.
2020	[24]	Presented an analysis for three FS methods evaluating their accuracy in order to find out the efficient method.
2020	[25]	Presented a comparative study to evaluate seven of the most commonly used and outstanding ranking based unsupervised FS methods.
2020	[26]	Performed a comparative study on 22 of state-of-the-art FS filter methods.
2021	[27]	Reviewed SI-based FS methods and provided a comparative analysis based on their performance.

focus on nature-inspired meta-heuristic methods and their role in the FS process. Authors in [24] presented an analysis for three FS methods evaluating their accuracy using C4.5 classification methods in order to find out the efficient method. Authors in [25] presented a comparative study to evaluate seven of the most commonly used and outstanding ranking-based unsupervised FS methods, which are under the category of filters. They have evaluated the methods based on the quality of selection and runtime. Authors in [26] performed a comparative study on 22 state-of-the-art filter-based FS methods. This article analyzes the FS filter methods based on runtime and prediction accuracy. Authors in [27] reviewed SI-based FS methods and provided a comparative analysis based on their performance. This survey partially focuses on SI-based FS methods. Table 1 shows a summary of recent research survey papers on FS techniques highlighting their contributions. Whereas, Table 2 illustrates the Comparative analysis of the existing surveys based on the category, evaluation process, search strategy, the underline structure, experimental analysis, quantitative analysis, and qualitative analysis .

It is clearly observed, from Table 2, there exist several surveys in the literature of FS for big data. Which have reviewed FS methods, but sill there is no single survey comprehensively, qualitatively, quantitatively, and experimentally covered and analyzed all categories of FS as we do in our taxonomy survey and analysis paper, we have filled all the aforesaid gaps and limitations, this article is distinct from the other currently existing surveys in that it is comprehensive and up-to-date and board. It presents a taxonomy of FS methods along with qualitative and quantitative analysis. The article covers all the categories of FS (traditional and online) and their sub-categories traditional (Filter, Wrapper, Embedded and Hybrid), online (individual and group). Moreover, the proposed taxonomy survey conducted an experimental analysis of 12 various prominent filter and wrapper methods to evaluate their performance in terms of (accuracy, precision, recall, F-measures, and the number of selected features). It mainly focuses on describing and analyzing the current existing FS methods based on their objective, search strategy, evolution process, learning task, feature structure, strength, and weaknesses. This survey is meant to improve the efficiency in fast learning of the state-of-the-art FS methods; moreover, it assists in finding out the limitations of current existing FS methods and potential research gaps. Definitely, this taxonomy will expand the researchers' minds and pave their way for possible new methods and future research directions.

3 Feature selection

Feature selection (FS) is a key aspect of data processing and machine learning methods. It is a primary task in the process of knowledge discovery. This process often faces the

Table 2 Comparative analysis of the existing surveys of FS

Survey	Year	Traditional Feature Selection						Online Feature Selection		Experimental analysis	Quantitative analysis	Qualitative analysis
		Filter		Wrapper		Embedded	Hybrid	Individual	Group			
		UN	MU	SSM	HSM							
[10]	2014	✓		✓	✓	✓				✓		
[17]	2014	✓	✓	✓		✓		✓				
[8]	2016	✓	✓							✓		
[11]	2017	✓	✓	✓		✓				✓		✓
[18]	2018	✓	✓	✓		✓		✓		✓		✓
[12]	2018	✓	✓	✓	✓	✓				✓		
[20]	2018	✓	✓	✓		✓		✓	✓	✓		
[15]	2018	✓	✓	✓		✓	✓					
[19]	2019	✓	✓		✓	✓	✓	✓				✓
[13]	2019	✓	✓	✓	✓	✓					✓	✓
[21]	2019	✓	✓	✓		✓		✓	✓			✓
[16]	2019	✓	✓	✓	✓	✓						✓
[14]	2020	✓	✓	✓		✓	✓			✓		✓
[22]	2020	✓	✓							✓		✓
[23]	2020				✓						✓	✓
[24]	2020	✓	✓	✓			✓			✓		
[25]	2020	✓	✓							✓		✓
[26]	2020	✓	✓	✓						✓		✓
[27]	2021	✓	✓	✓	✓	✓	✓			✓		✓

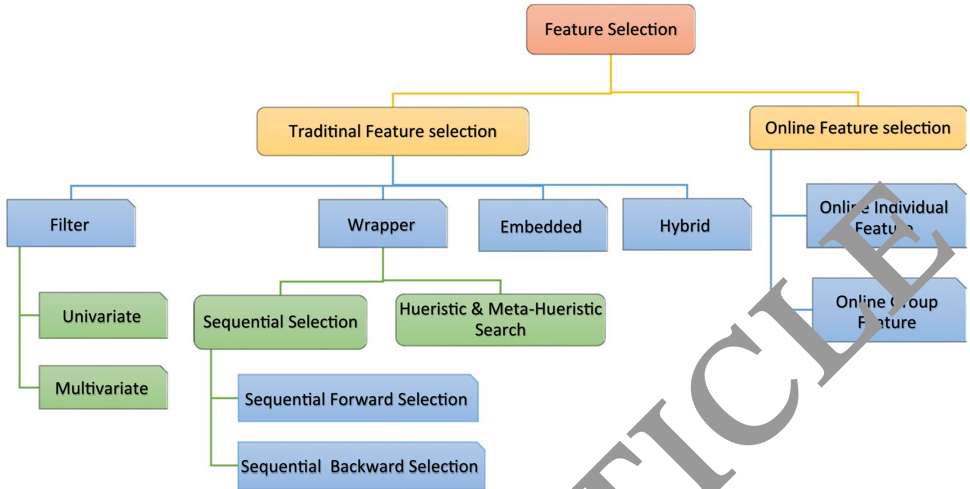
(UN Univariate, MU Multivariate, SSM Sequential Selection Method, HSM Heuristic Search Method)

problem of data high dimensionality, noisy and non-relevant features that exponentially increase the memory and time requirements for processing the data. FS is a key task for creating an effective and efficient learning model; it also helps in reducing the number of considered features while building learning models. FS improves the quality of data processing and makes the process of data modeling more efficient. FS methods provide many benefits such as reducing the storage requirements, increasing learning performance, improving the prediction accuracy, and leading to better comprehensibility. Thus, it speeds up the learning process with less memory consumption. Generally, the FS process consists of four steps:

- Generating the subset: a set of features is determined according to a specific searching strategy.
- Evaluating the subset: the selected candidate subset is evaluated based on an evaluating criterion.
- Stopping criterion: The set of features best suited for the evaluating criterion is selected from all the candidates evaluated after fulfilling the stopping criterion.
- Validating the results: The selected subset is validated by using domain knowledge or a test set.

Generally, the most significant steps in the process of feature selection are how to search for the optimum feature subset and evaluate the generated subset. To search for the optimal subset: sequential and heuristic-based search methods can be utilized. Whereas to evaluate the generated subset, the filters, wrappers, and embedded can be used. This survey study mainly focuses on reviewing and analyzing the searching strategies and evaluation methods. This section provides the taxonomy of FS methods, where it broadly classifies them into two main categories: Traditional Feature Selection (TFS) and Online Feature Selection (OFS). TFS implies that all the features are known in advance before the learning process starts, this implication is often non-practical in lots of real-world applications especially when training data is generated dynamically one-by-one over time. Therefore, in such a scenario, the TFS does not really perform well. Another interesting scenario assumes that the features are dynamically generated they also arrive individually or in groups and need to be processed upon their arrival that is known as OFS. TFS methods are further classified into three sub-categories: filter, wrapper, embedded and hybrid. On other hand, OFS methods are classified into online individual FS and online group FS as illustrates in Fig. 1.

Fig. 1 Taxonomy of feature selection methods.



3.1 Traditional Feature Selection (TFS)

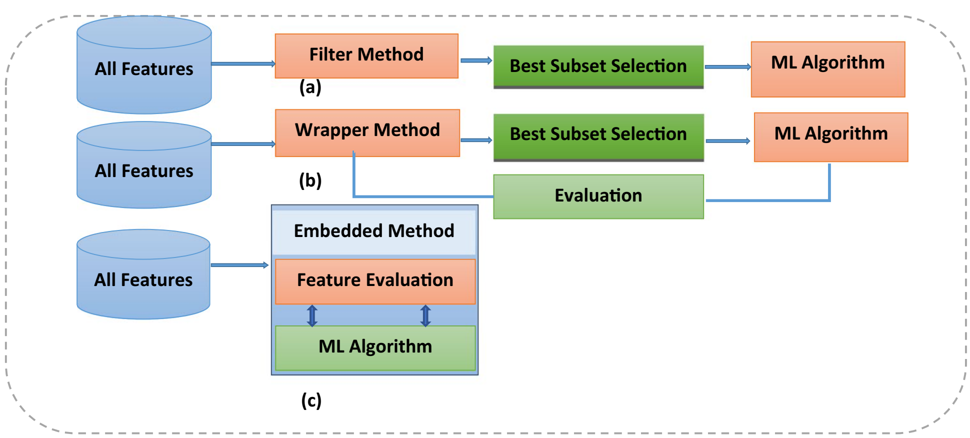
TFS techniques assume that the process of FS is performed in an offline fashion, where entire features are given to a learner in advance before the FS process takes place. This sub-section describes and analyzes the various traditional FS sub-categories: filter, wrapper, embedded and Hybrid method.

3.1.1 Filter method

Filter methods work based on different statistical characteristics of the data in order to compute feature relevance and perform the process of selecting features without taking into consideration the learning techniques. Figure 2(a) demonstrates the steps involved in the process of filter-based FS methods and their workflow. These methods apply statistical tests such as distance correlation and information to give a score for each feature based on its significance, then the subset of features with the highest score is selected to create a classifier and eliminate other features from the dataset. This type of methods has two phases: relevance ranking and evaluation. In the first phase, feature importance is ranked based on some measures such as correlation, consistency, distance, dependency, similarity, or information. The ranking process can be generally categorized into two approaches: univariate and multivariate. The univariate filter approach is also called feature ranking which assumes that every feature is independently ranked without considering dependencies between features, whereas the multivariate approach which is also known as the subset search method, ranks multiple features in a batch manner and evaluate the relevance of features with respect to their interdependence. Thus, a multivariate method is capable of dealing with redundant features. In the second phase of the typical filter technique, the higher-ranged features are selected to create a classifier, and the lower-ranked features are filtered out.

Univariate filter method The univariate approach is also known as the feature ranking-based technique. It works on assumption that each feature gets weighted by using one of the evaluation metrics such as statistical or information

Fig. 2 Feature Selection methods Filter, Wrapper, and Embedded.



measures. The features are ranked independently according to weights assigned to them regardless of other features. Then the top-ranked features get selected as the most important features based on a predefined threshold that defines the number of features to be chosen from a dataset. This sub-section briefly describes the existing univariate filter-based FS methods. The following are some univariate filter methods used for FS. Liu et al. [28] applied the chi-square statistic measure to weight the features and then arrange them accordingly to choose the features of greater importance, also some other recent studies have applied chi-square to select features in text classification [29–31].

Duda et al. [32] proposed Fisher Score (F-Score) which is a widely used supervised-based FS method that evaluates each feature individually to choose a small set of relevant features. It selects features based on their similarity, where similar features value of the sample is assigned to the same class, whereas other different features value of the sample is assigned to different classes. Then the fisher score is evaluated for every feature where the features with the highest score are chosen as relevant features. Fisher Score evaluates features independently; thus, it does not have the capability of dealing with the redundancy of features. Gu et al. [33] developed a generic Fisher Score to jointly choose features, it tends to find a subset of features, which increase the lower bound of the standard Fisher Score. However, Fisher Score ignores the association between features. To solve this issue, Gan et al. [34] proposed FS method, named iteratively local Fisher score (ILFS) in order to make the local within-class scatter smallest and the local between-class scatter greatest. Luo et al. [35] developed SVM technique based on Fisher-Score FS for behavior analysis of students to select optimal features related to learning behavior.

Rao et al. [36] developed an approach called PCA-Entropy to rank the features of an unlabeled dataset using the principal components based on the concept of representation entropy. Hall et al. [37] proposed the Information Gain (IG) based method, which is a simple univariate filter that measures the mutual information for every feature with the class label then produces an ordered ranking of all of the features based on their IG. Finally, selects the top-ranked features and remove others. To enhance the accuracy of heart attack, Ramesh et al. [38] proposed FS method based on IG in order to identify the most relevant features that help in determining the prediction of a heart attack. Pratiwi et al. [39] proposed FS method based on IG for document sentiment analysis in order to deal with a large number of features and obtained better sentiment classification.

He et al. [40] developed FS approach named Laplacian Score (LS) that ranks features based on their power of locality preserving, then the higher-ranked features get selected as the most important features. LS does not consider the association between features, Pang et al. [41] introduced an improved form of LS, named forward iterative LS (FILS) to overcome this issue. Witten et al. [42] introduced Symmetrical Uncertainty (SU), which is a correlation measure between both the features and the target class, which calculates the significance of the features based on information gained. Features with a high SU value get higher importance. SU takes values, which are normalized to the range [0, 1]. Moreover, some other researchers [43–48] utilized univariate filter to reduce the high dimensional text dataset. Rehman et al. [49] introduced a feature ranking method, called normalized difference measure (NDM) which takes into consideration the relative document frequencies. Patel et al. [50] proposed FS method for fault diagnosis based on Euclidean distance to select a significant feature subset for accurate diagnosis.

Multivariate filter method The multivariate approach, also known as a feature subset-based method that evaluates the quality of each feature concerning the other features by using either one of the statistical measures or supervised learning algorithms to identify the significance of each subset, and the most significant subset gets selected as the significant feature subset for a given dataset. This sub-section briefly describes the existing multivariate filter-based methods. Such methods include ReliefF which was proposed by Kira et al. [51]. In this method, each feature gets weighted depending on their importance with the target class, high weighted features get assigned to the same class whereas non-matching values get assigned to different classes. However, Relief can only deal with binary classification problems. The updated Relief version proposed by [52–54] are capable of managing multi-label classification problems. However, those methods are successful attribute estimators but are still not capable to remove the redundant features. To overcome the limitation of ReliefF family methods, Raj et al. [55] proposed a filter-Based FS named Boundary Margin Relief (BMR) algorithm to identify important features from high-dimensional redundant microarray data. Due to the uncertainty and noise of the features in the Relief methods, to address this issue, Munirathinam et al. [56] proposed a FS algorithm called noisy feature removal-Relief (NFR-Relief) to improve the classification performance.

Zhao et al. [57] proposed FS method called INTERACT to select relevant features. This method is implemented in two stages: initially, it ranks features according to a mutual information-based measure, then it individually evaluates the variables based on their consistency contribution, in such a way that only the features whose consistency contribution reaches a predefined threshold gets selected. Hall [58] proposed a multivariate filter method named Correlation-Based FS (CFS), which evaluates subsets features by choosing subsets of features, which are deeply correlated with the class but have low inter-correlation among each other. Dash et al.

[59] proposed a multivariate technique called Consistency-Based Filter (CBF) that measures the consistency with the class, then selects out features with high consistency and uses an inconsistency measure to determine feature to be removed. Liu et al. [60] developed a new method named Fast Correlation Based Filter (FCBF) that use a Symmetrical Uncertainty (SU) measure to evaluate the features' significance and find the correlation between both relevance and redundancy, then select the feature that highly correlated to the class but not highly correlated to any of the other features. Ding et al. [61] proposed a method named Minimal Redundancy and Maximum Relevance (mRMR) that select out the feature based on the feature significance and redundancy. The relevance is calculated by using mutual information and the redundancy of a feature, which is determined based on mutual dissimilarity to other features. Bennasar et al. [62] proposed two methods based on information theory to select important features from high dimensional data; the proposed method aims to overcome the problem of lack of interaction between features and classifiers. For text classification, Labani et al. [63] proposed a filter technique for FS, named Multivariate Relative Discrimination Criterion (MRDC). This method focuses on the reduction of redundant features using mRMR concepts, thus improves the classification performance. Sheikhi et al. [64] Proposed a multivariate filter method based on ranks of positive instances. In this method, redundancy is replaced by diversity to determine the relevance of a candidate feature compared to the already selected subset and improve the performance of the selection process.

To address the issue of a poor understanding of the imbalanced data, Shahee et al. [65] proposed a novel distance-based feature selection strategy (ED-Relief), which uses a complex distance metric to manage the phenomena of both between and within-class imbalance occurring at the same time. Du et al. [66] proposed FS method called joint imbalanced classification and FS (JICFS), which aims to pick important features and improve the classification model's performance. Whereas Hua et al. [67] utilized an approximate Markov blanket and proposed a filter-based FS algorithm that is suitable to high-dimensional datasets. Susan et al. [68] presented a mutual information with Gaussian gain for measuring the relationships between features and the class, as well as in-between features, then they proposed a filter approach for FS. In addition, Wu et al. [69] proposed FS method based on mutual information to select candidate features for the grading of glioma. For multi-label classification problem, Sun et al. [70] introduced FS method based on multi-label fuzzy neighborhood rough sets (MFNRS) and mRMR that can be used on multi-label data with missing labels. Furthermore, Qian et al. [71] based on the rough set, proposed FS algorithm for multi-label classification. This method obtained superior outcomes by conducting different importance of labels, Label distribution learning is utilized in such applications. Lin et al. [72] proposed multi-label FS based on neighborhood mutual information to choose optimal features from the multi-label dataset and improved the classification performance. Zhang et al. [73] analyzed various typical information-theoretic FS approaches, such as mRMR, JMI, and FOU, from the viewpoint of the lower boundaries of underlying correlations of the features. Qualitative analysis is illustrated in Table 3 which summarizes them based on their (object, schema, measure, and learning task).

3.1.2 Wrapper method

Another important category of FS is the wrapper method. It employs learning algorithms to evaluate candidate feature subsets to determine the importance of each feature subset and adopts a search method to choose the optimal final subset. Generally, the wrapper methods can be classified into two main categories: Sequential search and Heuristic and Meta-Heuristic search methods. Figure 2(b) shows the steps involved in the process of wrapper-based FS methods. Moreover, a qualitative analysis is illustrated in Table 4, which compares wrapper methods based on their (objective and search strategy).

Sequential selection methods The Sequential selection methods sequentially access the features from the given feature space, which applies a search procedure and generates the important subset of features. They can be divided into two types:

Sequential Forward Selection (SFS)
In SFS, the searching process initially starts with no features at all, and then sequentially keeps on adding a candidate feature that can improve the overall learning process until the best results are found and no more improvement in the evaluation function. One common method of forward-searching procedure is the SFS technique developed by Whitney [74] the main idea of this method is that it initially begins with an empty candidate features set, then it keeps adding the optimal feature that has the most important improvement in the classification performance of learning algorithms, it adds them one at a time until the wanted amount of features is satisfied or some stop conditions are met. The major issue in this method is that the dependency between the features is ignored.
Pudil et al. [75] developed an improved method of SFS called Sequential Floating Forward Selection (SFFS), which is much more elastic than SFS as it adds an extra step before each iteration, SFFS excludes a feature from the subset obtained and the subset in each iteration evaluates the new feature subsets. Both SFS and SFFS suffer

Table 3 summary of filter-based techniques for FS

Algorithm	Objective	Scheme	Measure	Applicable to task	Ref.
Chi-Squared	Statistically weight the features and rank them to select the most significant ones.	Univariate	Statistical	Classification	[28–31]
Fisher Score	It picks up features in such a way that the feature with identical instance values should be assigned to the same class while the feature with different instance values should be assigned to different classes.	Univariate	Statistical	Classification	[32–35]
PCA-Entropy	To rank the features of an unlabeled dataset using the principal components based on the concept of representation entropy.	Univariate	Information	Clustering	[36]
Information Gain	To calculate the mutual information for every attribute and class, then it generates an ordered ranking of all features.	Univariate	Information	Classification	[37–39]
Laplacian Score	To rank features based on their power of locality preserving, and then the higher-ranked features are selected.	Univariate	Similarity	Classification and Clustering	[40, 41]
Symmetrical Uncertainty (SU)	To evaluate the correlation between the features and the target class then measure the quality of the features on the basis of information gain. High-symmetric features are selected.	Univariate	Information	Classification	[42]
ReliefF and ReliefF	To randomly select an instance of the data and then find its closed neighbor from the same and opposite class.	Multivariate	Distance	Classification and Regression	[51, 52, 55, 56]
INTERACT	Selects the most significant features while exploring the interaction of the feature implicitly.	Multivariate	Information	Classification and Regression	[57]
Correlation-Based FS (CFS)	To identify the features which are highly correlated with the class but uncorrelated with each other.	Multivariate	Statistical	Classification and Regression	[58]
Consistency-based Filter	To obtain subsets of features, which are selected according to the consistency level with the class.	Multivariate	Consistency	Classification and Regression	[59]
Fast Correlation-Based Filter (FCBF)	To simultaneously exploit feature class correlation as well as feature-feature correlation. Initially, the method has given a predetermined threshold and it chooses a subset of strongly correlated features with a class label.	Multivariate	Information	Classification and Regression	[60]
Minimum Redundancy, Maximum Relevance (mRmR)	Selecting the feature based on relevance and redundancy.	Multivariate	Information	Classification and Regression	[61]

Table 4 Summary of wrapper-based techniques for FS.

Algorithm	Objective	Search strategy	Reference
Sequential Forward Selection (SFS)	Sequentially adds a candidate feature one at a time that has the most significant improvement for the learning algorithm classification performance.	Sequential Search	[74, 79, 80, 82]
Sequential Backward Selection (SBS)	It works in the opposite direction of sequential forward selection, starts with the full set, and removes one most irrelevant feature, which gives the lowest decrease in classification accuracy in each iteration.	Sequential Search	[81]
Sequential Floating Forward And Backward Selection (SFFS and SFBS)	Adds step before each iteration, SFFS excludes a feature from the subsets obtained, and subsets in each iteration evaluate the new feature subsets.	Sequential Search	[75]
Adaptive Sequential Forward Floating Selection (ASFFS)	Attempts to overcome the nesting effect. By searching for a subset in both directions.	Sequential Search	[76]
Genetic Algorithm (GA)	To produce the feature subsets and the supervised machine-learning algorithm is used for evaluating the generated subset.	Heuristic Search	[104–114]
Particle Swarm Optimization (PSO)	To improve the selection process of the final feature subset.	Heuristic Search	[83–86, 115–126]
Ant Colony Optimization (ACO)	To identify features that are strongly correlated with the class but uncorrelated with each other.	Heuristic Search	[87, 127–134]
Gray Wolf Optimization (GWO)	To imitate the natural hunting behavior of a pack of grey wolves and optimize the selection process.	Heuristic Search	[88–90]
Salp Swarm Algorithm (SSA)	Proposed as a way to enhance FS accuracy. This method inspired by the swarming behavior of salps while migrating and foraging in the seas technique.	Heuristic Search	[91–96]
Artificial Bee Colony optimization (ABC)	To mimic the behavior of bees in their quest for food in order to solve difficult optimization issues and improve the FS task.	Heuristic Search	[97, 98]
Bat Algorithm (BA)	To improve the performance of FS. This method is inspired by the collective behavior of bats in the natural environment.	Heuristic Search	[99]
Firefly Algorithm (FA)	To increase the accuracy of FS. This method inspired by optical connectivity between fireflies.	Heuristic Search	[100]
Whale Optimization Algorithm (WOA)	To simulate humpback whale's hunt for prey, encirclement prey, and bubble-net foraging behavior in order to improve the selection mechanism.	Heuristic Search	[101]
Cuckoo Search (CS)	To increase the performance of FS task. This method inspired by the peculiar habits of a bird known as the cuckoo.	Heuristic Search	[102]
Simulated Annealing (SA)	To simultaneously exploit feature-class correlation as well as feature-feature correlation.	Heuristic Search	[135–137]
Tabu Searching (TS)	To solve the combinatorial optimization problem where an optimal ordering and selection of options is desired	Heuristic Search	[138–140]
Genetic Programming (GP)	To obtain subsets of features according to the level of consistency with the class.	Heuristic Search	[141–144]

from nested subset drawbacks, which means that, cannot deal with feature redundant problems, because all these methods are irrevocable. To overcome this issue and save computing time, an adaptive form of the SFFS was developed by Somol, et al. [76], the Adaptive Sequential Forward Floating Selection (ASFFS) technique aims to select fewer redundant feature subsets by searching for a subset in both directions either adding S features or removing R features at each step. It uses parameters S and R to adaptively control the number of added and removed features respectively. It uses a parameter S to define the number of features that have been calculated adaptively to be added in the inclusion phase. In the exclusion phase, the parameter R is used to remove the maximum number of features that lead to improved performance.

Some other method also aims at solving the nesting issue, such as the Plus-L-Minus-r search methods, Nakariyakul et al. [77] proposed an Improved Forward Floating Selection (IFFS) method that enhances the SFFS algorithm. It adds an extra searching step in which a weak feature is replaced in each step to form a new and sufficient subset by checking whether the current feature set can be enhanced by eliminating any feature in the chosen subset of features or adding a new one.

Sequential Backward Selection (SBS)

Marill et al. [78] also proposed SBS, which works in the reverse direction of sequential forward selection begins with the entire features and proceeds to eliminate most irrelevant features at every step, whose elimination gives the best result on classification accuracy. The elimination procedure stops until a wanted amount of features are satisfied or some stop conditions standard is met. Yan et al. [79] proposed sequential FS method and cost-sensitive for the chiller to select the most significant features. Aggrawal et al. [80] proposed a sequential FS algorithm to choose the most important features in heart disease. Moreover, Ruan et al. [81] suggested SBS based feature selection method composed with RF classifier to increase the detection accuracy of sulfur and phosphorus in steel. Gong et al. [82] introduced a sequential selection method to select features in high-dimensional datasets and enhances selection accuracy.

Heuristic search algorithms This section describes another category of wrapper methods that work based on the search strategy. In practice, the heuristic function is applied to get the prior knowledge for directing the search process, generate the subsets, and then evaluate these subsets utilizing the classification algorithm. Heuristics-based search can be utilized to test all available feature subsets by exploring the entire search space repeatedly to speed up the selection process and get the optimal solution. Since the sequential methods typically evaluate all possible solutions and then choose the optimal subset of features, it is computationally complex, specifically when the feature dimensionality is high. To overcome the limitations of sequential methods, heuristic search-based metaheuristic algorithms are adapted such as Particle Swarm Optimization (PSO) based on a single objective [83–85] based on multi-objective (MOPSO) [86], Ant Colony Optimization (ACO) [87], Gray Wolf Optimization (GWO) [88–90], Salp Swarm Algorithm (SSA) [91–96], Artificial Bee Colony optimization (ABC) [97, 98], Bat Algorithm (BA) [99], Firefly Algorithm (FA) [100], Whale Optimization Algorithm (WOA) [101], Cuckoo Search (CS) [102], Harris Hawks optimizer (HHO) [103], Genetic algorithm (GA), Simulated Annealing (SA), Tabu Searching (TS), and Genetic programming. These algorithms have the ability to perform a global search, therefore, they have shown promising results in solving single and multi-objective optimization problems.

GA-based Methods: GA can work in combination with other algorithms to improve the learning process and accuracy, such as supervised machine learning, where the GA can be applied to generate a feature subset and the supervised machine learning algorithms for evaluating the generated subsets. Oreski et al. [104] presented FS method based on GA and NN (Neural Networks) for credit risk assessment. Whereas Kari et al. [105] proposed a hybrid FS approach by applying a GA with a combination of Support Vector Machine (SVM) to improve fault diagnosis accuracy. In addition, Welikala et al. [106] developed FS method by using a GA to generate the subsets along with SVM to evaluate the generated subsets for mining the medical dataset. Jiang et al. [107] presented a method for FS applying SU together with GA based on the Naïve Bayes classifier. Lu et al. [108] presented a method for FS based on a GA to classify electroencephalogram (EEG) patterns that aim to select optimal feature subsets.

Wang et al. [109] developed FS method using the GA to generate feature subsets with SVM to evaluate the generated feature subsets and obtain an optimal result to be used in data classification applications. Khammassi et al. [110] applied a combination of the GA with Logistic Regression (GA-LR) to detect the intruders in the network. The heuristic search strategy utilizing Genetic Alg. has generated the best optimal subset of features. Erguzel et al. [111] introduced a method for FS by applying a combination of Artificial Neural Network (ANN) with GA to classify the EEG signal. Li et al. [112] presented the FS approach using a GA with SVM to be used in image classification applications. Das et al. [113] developed a technique to select significant features by applying a GA with SVM to be used in handwritten digit recognition applications. Maleki et al. [114] utilized GA to efficiently select features for reducing the high dimensionality datasets with KNN classifier for enhancing the classification process for patients' disease diagnosing.

PSO-based Methods: PSO is also a heuristic-based optimization technique, which is utilized to produce feature subsets and validate them using a supervised algorithm to define the important feature subset. Wang et al. [115] developed a new FS technique based on PSO to choose the most optimal feature subsets and apply the rough set reduction. Y. Zhang et al. [116] developed a PSO-based multi-objective FS method named a Hybrid Mutation with Multi-Objective PSO (HMPSOFS) for the cost-based FS issue that improves the classification performance at a low cost. Xue et al. [117] study the multi-objective PSO for FS and proposed two PSO-based multi-objective FS methods for classification. The first method applies the idea of non-dominated sorting into PSO to address issues of FS, this method is called Non-Dominated Sorting into PSO FS (NSPSOFS). The second method employs the ideas of Crowding, Mutation, and Dominance to PSO for FS (CMDPSOFS), which searches for the Pareto front solutions. Jain et al. [118] presented a hybrid model for gene selection and cancer classification which combines Correlation-based FS (CFS) with Improved Binary PSO (iBPSO) to choose a low-dimensional dataset of genes for single and multi-class cancers and to enhance the performance of classification.

Chen et al. [119] proposed FS method using PSO search for a sleep disorder diagnosis system. Yang et al. [120] presented PSO-based FS for land cover classification. Xue et al. [121] developed a PSO-based FS method to select important features and also improves the performance of classification. Chen et al. [122] developed a wrapper-based approach, which combines PSO and a spiral-shaped mechanism to select the relevant features. Dhrif et al. [123] proposed a novel hybrid wrapper and PSO-based filter algorithm, which aims to find the smallest and most stable feature subsets, eliminating redundant and non-relevant features. Duarteet al. [124] introduced a Hybrid PSO with a Spiral-Shaped Mechanism (HPSO-SSM) that seeks to select optimal feature subset and enhance classification accuracy. Xue et al. [125] suggested FS method based on self-adaptive parameters and PSO (SPS-PSO) by employing multiple classifiers to address the scalability of FS problem. Zhou et al. [126] introduced an enhanced discretization-based PSO for high dimensional data that aims for reducing the number of features.

***ACO-based Method*:** Several FS methods have employed the ACO as the search criteria to generate a subset, such as Sivagaminathan et al. [127] developed FS method for the medical diagnosis system to select optimal features by applying ACO-based FS with ANN. Aghdam et al. [128] develop an optimized-based FS method for text classification application by applying ACO to generate feature subsets and then employ the nearest neighbor classifier to validate the result. Ahmed et al. [129] suggested a novel text FS method, which utilizes a wrapper approach, combined with ACO to resolve the FS issue. It utilizes the K-Nearest Neighbor (KNN) as a classifier for evaluating and generating an optimal candidate feature subset. Besides, the researcher Kanan et al. [130] has applied the same ACO as FS method for the face recognition system. In the suggested work, the K-NN classifier is adopted for testing the derived subset by ACO as FS method. Jayaprakash et al. [131] have proposed FS approach by using ACO algorithm to define the most important and definitive features. Then, these features are fed into the ensemble SVM to enable both roadside traffic detection and recognition. Ghosh et al. [132] have developed a multi-objective FS based on a wrapper-filter combination of ACO, they used a filter method alternative of utilizing a wrapper method for subset evaluation and to reduce computational complexity. This method was developed for solving the traveling salesman problem. Tabakhi et al. [133] proposed new multivariate FS methods based on the ACO technique by considering the redundancy and relevance of features. Moradi et al. [134] presented a filter-based FS technique using the graph clustering method and ACO for classification problems.

SA-based Methods: SA is also adapted to generate the feature subset for evaluations, such as Lin et al. [135] employed the SA search to produce the feature subsets and evaluate them by a supervised learning algorithm called Back-Propagation Network (BPN) that selects optimal feature subsets. Meiri et al. [136] have employed SA-based FS to choose an optimal feature for marketing applications. Lin et al. [137] used a combination of the SA with SVM for FS. In the existing.

TS-based Methods: Many FS methods have used the Tabu Search optimization as the searching criteria for subset generation. We briefly describe some of such methods as follows. Wang et al. [138] proposed a novel approach for FS based on Rough set and Tabu search (FSRT), for credit scoring. Yan et al. [139] proposed a new FS method based on the Binary Chemical Reaction Optimization algorithm (BCRO) and Tabu Search is integrated with CRO to enhance search capacity. K-Nearest Neighbours (KNN) classifier is used for evaluating the quality of the chosen candidate subset. Chuang et al. [140] developed a hybrid method based on the Tabu Search combined with binary PSO to solve the FS problem.

GP-based Methods: Other FS methods also used SA to generate the feature subsets, such as the FS method proposed by Chen et al. [141] which used permutation to choose optimal features for high dimensional Symbolic Regression (SR). Mei et al. [142] developed a new FS method for job shop scheduling based on GP. Neshatian et al. [143] proposed FS algorithm based on GP for classification tasks. They designed a function to measure the relevance of subsets of features. Sandin et al. [144] developed a GP-based FS method to deal with the unbalanced high-dimensional classification tasks, which has combined several commonly used FS metrics to improve the classification performance.

3.1.3 Embedded methods

The embedded method is a mechanism that is integrated with a classifier for guiding the process of FS, as shown in Fig. 2(c). The main idea is to include the process of FS in the interaction with learning models and use its characteristics to handle the evaluation of features. The advantage of embedded methods is that they interact with the classifier, and have a lower computation complexity than wrappers. Many research works have been proposed utilizing these

Table 5 Summary of embedded-based techniques for FS.

Algorithm	Objective	Type	Reference
Least Absolute Shrinkage and Selection Operator (LASSO)	Is a regression analysis method that changes coefficient estimation and makes some of them zero to perform FS as it uses the nonzero ones as the selected features.	Supervised	[145]
Relaxed Lasso and Generalized Multi-Class Support Vector Machine (RL-GenSVM)	An upgraded version of LASSO To deal with the problem of multi-label classification.	Supervised	[146]
Enumerate Lasso	This method provides a solution to the Lasso regression problem for enumerating search.	Supervised	[147]
HSIC Lasso	Used to perform non-linear regression tasks and select the optimal feature from high dimensional data sets.	Supervised	[148]
Sparse Proximal Support Vector Machines (sPSVMs)	To do FS of High Dimension Low Sample Size (HDLSS) datasets by incorporating sparsity in Proximal Support Vector Machines (PSVMs).	Supervised	[149]
locality and similarity preserving embedding (LSPE)	To select significant features based on locality and similarity preserving.	Unsupervised	[150]
Recursive Feature Elimination for Support Vector Machines (SVM-RFE)	It is employed SVM as a classifier, then chooses features and removes those features that get least important based on the weights result from the SVM classifier	Supervised	[151]
variable step size (VSSRFE)	Is an improved version of RFE, which tries to reduce SVM-RFE time consumption.	Supervised	[152]

methods, this sub-section briefly describes the state-of-the-art of the existing embedded FS methods, moreover, a qualitative analysis is illustrated in Table 5 which compares them based on (objective and type).

Tibshirani. [145] proposed a method named Least Absolute Shrinkage and Selection Operator (LASSO) for gradient in linear models. This method works based on a regression analysis strategy that changes coefficient estimation of a linear classifier and makes some of the feature coefficients zero to perform FS. It uses the nonzero as the selected features. Further, Kang et al. [146] proposed an upgraded version of LASSO to deal with the problem of multi-label classification, this method is called Relaxed Lasso and Generalized Multi-Class SVM (RL-GenSVM). Similarly, Hara et al. [147] proposed solutions for the LASSO regression problem to enumerate search. The LASSO assumes that a linear correlation between features and the target class exists, but it does not work well in the non-linear scenario. In addition, Makoto et al. [148] proposed an upgraded method of LASSO used to perform non-linear regression tasks to select an optimal feature from high dimensional data sets by considering a feature-wise kernelized Lasso for picking up a non-linear input-output feature values dependency. Then non-redundant features with a high statistical dependence on output values can be found in terms of kernel-based independence measures such as the Hilbert-Schmidt Independence Criterion (HSIC Lasso). Finally, the overall optimum solution can be calculated efficiently.

Pappu et al. [149] proposed a novel embedded FS technique named Sparse Proximal SVM (sPSVMs), by combining sparsity with Proximal SVM, to perform the FS of High Dimension Low Sample Size (HDLSS) datasets. Fang et al. [150] proposed an unsupervised FS technique that aims to choose an optimal feature based on Locality and Similarity Preserving Embedding (LSPE). Guyon et al. [151] developed a novel technique called SVM for Recursive Feature Elimination (SVM-RFE), which is an embedded mechanism that employed SVM as a classifier, then chooses features and removes the least significant features based on the weights result from the SVM classifier. The major disadvantage of SVM-RFE is time consumption. However, to overcome this drawback, Zifa et al. [152] introduced an upgraded method of RFE named Variable Step Size with RFE (VSSRFE) that attempts to reduce SVM-RFE time consumption by adding a large step size and makes the step size keep decreasing whereas the number of features to be removed is decreased.

Guha et al. [153] an embedded version of WOA named embedded chaotic whale survival algorithm (ECWSA) has been proposed to improve classification accuracy and a select subset of features with low computation cost. Shankar et al. [154] introduced an embedded-based weighted FS method for classifying web documents. You et al. [155] proposed an embedded FS for the multi-label classification of music emotions. Recently, embedded FS techniques have been proposed [156–164]. Imani et al. [165] introduced an embedded method based on a Chi-square in the filter step, and then the ACO and GA were utilized to select significant features in the wrapper step.

3.1.4 Hybrid method

The hybrid method combines both filter and wrapper in order to utilize the advantage of both methods. The filter technique is used to obtain a good subset, and the wrapper technique is later used to enhance the result. Recently, researchers have proposed some hybrid FS methods, which will be briefly described in this section.

Inbarani et al. [166] proposed a hybrid FS method by combining the wrapper PSO and rough sets theory to increase the classification accuracy of disease diagnosis. Also, Pashaei et al. [167] proposed FS method by utilizing a binary black hole algorithm and improved BPSO for cancer classification. Whereas, Kabir et al. [168] developed ACO-based hybrid FS method combining the neural network and IG technique to select relevant features. Besides, Selvakumar et al. [169] introduced the FA-based FS method by combining of filter and wrapper-based FS method to select the best features for intrusion detection. Furthermore, Al-Betar et al. [170] utilize the bat algorithm to search for sufficient features subset. This method integrated the filter and wrapper method to improve the performance of the selection process. Emary et al. [171] proposed Multi-Objective GWO to find optimal feature subsets. This method utilized a filter model for computation efficiency and wrapper to improve classification accuracy. Ibrahim et al. [172] introduced a hybrid optimization method for FS combining the slap swarm algorithm with particle swarm optimization to improve exploitation and the exploration steps. Furthermore, Tawhid et al. [173] a hybrid method of bat algorithm and improved PSO technique is proposed to improve the FS performance.

Lee et al. [174] proposed a hybrid method for FS that combined GA with Dynamic Parameter setting (GADP) for Microarray data analysis. Whereas, Salem et al. [175] presented a hybrid FS method that integrated IG and Standard GA for genes selection. Sharbaf et al. [176] hybrid method that uses Cellular Learning Automata and ACO was developed for gene selection. Alshamlan et al. [177] proposed a hybrid gene selection method that used mRMR with ABC algorithm called mRMR-ABC. Xie et al. [178] developed two-stage hybrid FS methods for the diagnosis of the erythemato-squamous disease. Sadeghian et al. [179] introduced a hybrid FS technique based on information theory and binary butterfly optimization algorithm to address the problem of selecting features from high dimensional datasets. Moreover, Amini et al. [180] proposed a hybrid method, which combines wrapper and embedded methods, GA adopted as wrapper and Elastic Net as an embedded method to increase the prediction accuracy of regression problems.

Song et al. [181] developed a hybrid FS method based on bare bones PSO (BBPSO) with mutual information to improve the FS process. Lu et al. [182] proposed a hybrid FS method that integrated the Adaptive Genetic Algorithm and Mutual Information Maximization called MIMAGA for gene expression selection. Moslehi et al. [183] proposed a hybrid filter-wrapper approach for FS that integrated GA and PSO in order to improve general classification task performance and Song et al. [184] proposed hybrid FS combining correlation-guided clustering and particle swarm optimization to select features from a high dimensional dataset. Further, El-Hasnony et al. [185] proposed FS technique based on the integration of grey wolf optimization and PSO to reduce the difficulties of FS process. For parallel FS, Abualigah et al. [186] proposed a parallel hybrid FS technique to tackle the problem of high-dimensional features in the text clustering application. However, to cope with the problem of multi-label FS, Hatami et al. [187] suggested a multi-label FS strategy that employs the mutual information measure to identify feature similarity and assesses the relevance of each feature with a number of labels, then employs ACO to rank the features according to on their relevancies. Furthermore, Zhang et al. [188] propose a multi-label FS approach based on global relevance and redundancy optimization. They provided a global optimization model that takes into account feature relevance and redundancy, making it easier to choose multi-label features. Hammami et al. [189] introduced a multi-objective hybrid technique for FS that combined GA and local search with the goal of reducing feature dimensionality and improving classification performance. To this end, a qualitative analysis is illustrated in Table 6, which compares (filter, wrapper, and embedded methods) based on their strengths and weakness.

3.2 Online feature selection (OFS)

The methods discussed earlier in the previous sections assume that all features and instances of data are available in advance before the learning process takes place. However, another interesting scenario assumes that features are produced dynamically and arriving one after another or group after group; they need to be processed upon their arrival, such a scenario called Online Feature Selection (OFS), which is more challenging than the TFS. In numerous real-world applications, there are two types of streaming feature data streams and feature streams. A primary distinction between data streams and the feature stream. In OFS the data stream, the number of features is fixed and the number of candidate instances is changeable over time. Furthermore, OFS with feature stream supposes that the number of instances of data is fixed whereas the number of candidate features arrives one at a time. OFS methods can be classified into to sub-categories: Online Individual Feature Selection (OIFS) and Online Group Feature Selection (OGFS). In the following sub-sections, we initially describe OIFS and review its existing related works, then we discuss OGFS with its existing related works. Finally, we provide a comparative analysis of these methods.

Table 6 Comparison of feature selection methods

Strategy	Advantage	Disadvantage
Filter	Efficient and computationally faster.	Ignores the correlation between classifiers.
	Independent of the learning algorithm.	It does not consider the correlation between the features.
	Faster than the wrapper and embedded methods.	Fails to recognize the patterns properly during the learning phase.
	Simple and suitable for low-dimensional data.	
Wrapper	Simple.	Possibility of overfitting.
	Interacts with the classifier.	Classifier dependent selection.
	Consider dependence between features.	Computationally intensive.
	Accurate more than filter method.	
Embedded	Interacts with the classifier	Classifier dependent selection.
	Accuracy is better than the filter method.	High computation cost.
	Computational complexity is better than the wrapper	
	Consider the dependencies between features.	

3.2.1 Online individual feature selection

Online individual feature selection shares the common assumption that candidate features are generated dynamically and arrive one at a time. This method is of high significance when dealing with real-world data-intensive applications, which require an efficient OFS method in order to cope with real-time data streaming applications, as a result, several studies have been proposed for OFS in the literature, this sub-section briefly describes the OIFS, moreover, in Table 7 a qualitative analysis is provided, which compares the exiting OIFS based on their objective and the limitations.

Perkins et al. [190] proposed one of the earliest techniques called Grafting to perform OFS to define a subset of online features that dynamically generate features one at a time. Grafting considers the selection of appropriate features as an integrated part of a prediction model within a regularized system. To train the predictive model, it employs stagewise logistic regression that progressively and gradually constructs a selected subset of features. The proposed method aims to pick a subset of features and return appropriate features at each step that must be as excellent as possible among the features seen so far. Grafting is efficient, and can also be used for both regression and classification. The main disadvantage of this method needs knowledge about the complete space of features and cannot deal with features of an unspecified number of features.

Zhou et al. [191] studied the problem of OFS in an incremental fashion where the size of feature space is unspecified in advanced and introduced Alpha-investing algorithms based on streamwise regression. This technique is one of the penalized probability ratios where the global model is not needed. The fundamental principle is that features are produced dynamically and a threshold of error reduction called the P-value needed for a new feature to join the model is dynamically adjusted. The P-value is introduced to decide either feature can be included in the model or ignored. However, one of the drawbacks associated with this method is that it does not evaluate the redundancy of chosen feature set which will greatly influence the subsequent selection process and it needs prior knowledge of the structure of the feature space to control the selection of significant features. Therefore, to tackle these drawbacks, Wu et al. [192, 193] discussed an interesting OSFS concept, in which the number of training data is unspecified and not all features are accessible for learning, while the size of the training samples is constant. Under this scenario, an OSFS framework with two approaches called OSFS and with a faster version, called fast-OSFS has been developed. The key idea is to pick features based on their relevancy and whether or not they are redundant. The proposed algorithms are capable of selecting appropriate features and eliminating redundant ones in real-time. However, when the dimensionality is incredibly high, such algorithms may still suffer computational cost complexities. Therefore, to address these drawbacks, Yu et al. [194] proposed a novel algorithm called SAOLA to handle the FS problem with exceptionally high dimensionality for large-scale data analytics. In order to effectively and successfully filter out redundant features, this algorithm employs theoretical analysis to derive minimal bounds of associations between features. SAOLA includes an online pairwise comparison technique with pair-wise comparisons to maintain relevant features in an online fashion over time. Instead of conditioning on a set of features, the SAOLA algorithm performs a set of pairwise comparisons among individual features. This tends to decrease the cost of processing.

Wang et al. [195] considered the problem of quickly increasing attributes in RS-theory for many real data sets and developed a novel technique for reducing incremental feature dimensions called DIA-RED. This strategy is based on data entropy for a dataset with dynamically growing features. The basic principle is that it employs the RS-based entropy

Table 7 Summary of OIFS methods

Technique	Objective	limitations	Ref.
Grafting	Grafting considers the choice of suitable features as an integral process of a prediction model in a regularized risk system based on a stagewise gradient descent approach. Which seeks to choose a subset of features and return features that are relevant.	Cannot deal with features of the unspecified number of features. Not ideal as some features dropped during the selection procedure. Suffer from nesting effect problem.	[190]
Alpha investing	To dynamically adjust a value of error reduction threshold called the P-value that is needed for a new feature to join the model. The P-value is introduced to decide whether the feature is to be added to the model or ignored.	Requires some prior knowledge. Redundancy never evaluated. Limited prediction accuracy.	[191]
OSFS and Fast-OSFS	An input feature can be divided into four basic distinct categories namely, highly relevant, poorly relevant but non-redundant, and unrelated. The proposed algorithms are capable of selecting strongly relevant and remove redundant ones in real-time.	Computational cost increases if the dimensionality is extremely high. Its runtime increased as the number of non-redundant features increases.	[192, 193]
SAOLA	Employing theoretical theory to control the selection of features with incredibly high dimensional space and derive a lower bound of associations for pairwise comparisons among features to eliminate redundancy.	The relevance threshold influences the performance It is difficult to obtain the optimum value for the threshold.	[194]
DIA-RED	To employs, the RS-based entropy value of the newly chosen subsets of features and upgrade that value, as new conditional features are included in the dataset.	Redundancy never evaluated. Does not apply efficiently to real-world data sets.	[195]
APSOFS	To overcome the incremental computation and to reduce the processing time in the classification model based on swarm search.	Computational cost increases as data increases.	[196]
FRSA-IFS-HIS	To select a significant feature using a fuzzy-RS model on a hybrid information system.	Does not scale well when the dimensionality is extremely high.	[197]
MLFS	The key idea is to weight each feature and obtain a features weights rank list based on the mRMR of every recently arrived label. After that, an optimization function is introduced to achieve the final feature ranking.	The size of the features is fixed in advance.	[198]
OS-NRRSAR-SA	Employing classical relevance analysis principles-based RS theory and removing irrelevant features in the OSFS problem.	Numerical features cannot be handled directly.	[199]
OSFSMI and OSFSMI-k	Applies the mutual information principle to calculate the correlation between features and to evaluate both redundancy and relevancy of features for classification tasks in an online fashion.	Computational cost increases as data increases.	[200]
OFS-A3M	This approach is based on a new neighborhood RS relationship with adjusted neighbors to pick important features in an online manner.	For actual datasets, it is too inflexible and leads to some redundancies in the selected subset of features.	[201]
OFS-Density	This method is based on an adaptive neighborhood relationship that uses the density information of the neighboring instances that will dynamically pick a correct number of neighbors during the FS process.	Not fast and it is not time efficient.	[202]
OSFASW	The main idea is that to use a self-adaptation sliding-window sampling technique and a redundancy analysis to minimize the dimensionality of the streaming features by eliminating the redundant and irrelevant features in real-time.	It is not time efficient.	[203]

Table 7 (continued)

Technique	Objective	limitations	Ref.
OFS	OFS aims to pick a limited number of online features for binary classification. OFS addresses two separate OFS tasks: (1) OFS by complete learning input and (2) OFS by partial learning input.	Do not remove redundant features.	[204]
MOANOFS	It is an intelligent OFS method based on automated negotiation using multiple collaborative learners to improve the FS system and enhance the decision-making system.	Not suitable for multi-classification and nonlinear domains.	[205]
OSFS-KW	To pick a small subset of the most significant features based on weighted dependency degree taking into account the neighborhood feature structure.	It is slower than some other methods.	[207]
SFS-RS	Considered the problem of SFS from the perspective of the RS, in which RS theory is used to handle the unidentified feature space.	Cannot handle numerical features directly.	[199]
KFOHFS	Used the kernelized fuzzy RS model that can successfully evaluate the fuzzy relation among features under the hierarchical label space.	Limited to a tree class structure.	[208]

value of the newly chosen subsets and upgrade that value as new conditional features are applied to the dataset. Regardless of having little or no knowledge of feature space, the DIA-RED deals with streaming environments, using only the information located in the lower approximation of a set and neglecting the information within the outer region. Hence, in real-world data sets, this technique does not always perform efficiently. Besides, an efficient mechanism that eliminates redundant features did not implement by the DIA-RED algorithm, which leads to the choice of a large subset of streaming features. Great computing usage may inevitably lead to incremental search space for data streams with high-dimensional data, especially in an environment of rapid change. With respect to this situation, Fong et al. [196] developed a new OFS optimization-based heuristic search called Accelerated PSOFSS (APSOFS). In particular, the proposed algorithm is intended to pick features based on swarm search upon its arrival that helps in achieving improved analytical performance within an appropriate processing time and find the correct combination of classification techniques.

Zeng et al. [197] defined the Fuzzy-RS (FRS) that is an efficient technique for FS in Hybrid Information System (HIS) and developed an Incremental FS approach called FRSA-IFS-HIS using a Fuzzy-RS model in HIS. The incremental technique is an efficient solution to handle FS in a complex and dynamic environment. They also introduced a novel Hybrid Distance (HD) that could work with various data types. The HD distance is combined with the Gaussian kernel and FRS. Lin et al. [198] investigated the interesting problem of multi-label FS with streaming labels and developed a new Multi-Label FS (MLFS) model in a dynamic label stream where the sample size of labels is unspecified in advance and the size of the features set is fixed. In this problem, the labels start showing up one at a time and when a new label arrives, the learning task is to evaluate features incrementally. In this situation, they initially, proposed an optimization system where the weight of the feature rank list of each label and individual feature rank list weights are obtained based on mRMR for every label that has recently arrived. After that, an optimization technique is added to reach the optimum feature rank list based on the specified weight values, the distance between each feature rank list and the final feature rank list is reduced, and then pick a feature list that minimizes the distance.

Eskandari and Javidi [209] considered the OSFS problem from the RS point of perspective and introduced a new OSFS approach known OS-NRRSAR-SA for OFS. The key benefit of the proposed approach is that data analysis based on RS does not require any domain knowledge other than the dataset provided. In RS theory, the proposed approach employs classical relevance analysis principles to manage an unspecified space of features and to remove irrelevant features in the OSFS problem that do not affect determining the final feature subset.

Rahmaninia et al. [200] proposed two novel filter-based FS approaches for the OSFS problem, called OSFSMI and OSFSMI-k to choose the most useful features from OSF. The proposed approaches used the mutual information principle to calculate the correlation between features and to evaluate both redundancy and relevancy of features for classification tasks in an online fashion. The proposed approaches seek to choose the most informative and non-redundant set of features that are highly relevant to the target class and to remove redundant and insignificant features. There are two key stages to the search process of the proposed methods. The importance of each newly arrived feature is determined in the first stage and those of strictly non-relevant features are removed. In the second stage, the usefulness of the related features is calculated by using both the principles of relevance and redundancy over multiple iterations.

Zhou et al. [201] defined a new neighborhood RS relation with adjusted neighbors that aims to automatically choose neighbor features that can achieve high serviceability, then use the information of adjusted numbers of neighboring instances around the target object and come up with a new OSFS method based on this relationship called OFS-A3M to perform FS in an online environment. This method included three evaluation measures "maximal-relevance, maximal-significance, and maximal-dependency". With the evaluation criteria, OFS-A3M can choose features with high relevancy, biggest dependency, and low redundancy. This method does not require any parameters to be specified in advance and does not demand any domain knowledge. Zhou et al. [202] developed a new OSFS approach based on the neighboring relationship of adapted density, called OFS-Density. More clearly, with the RS of the neighborhood, OFS-Density does not really involve knowledge of domain before actually learning. Meanwhile, introduced a new adaptive neighborhood relationship use the density information of the neighboring instances that will dynamically pick a correct number of neighbors during the FS process. OFS-Density can pick features with low redundancy using the fuzzy equal constraint.

Deng et al. [203] considered the OFS problem and developed a novel OFS method called OSFAW. The key idea is to use a self-adaptation sliding-window sampling technique and a redundancy analysis to minimize the dimensionality of the streaming features by eliminating the redundant and irrelevant features in real-time. An estimated Markov Blanket model with high prediction precision and a low number of features is provided by the proposed OSFASW.

Wang et al. [204] discussed the problem of OFS and proposed a new technique aimed at selecting a limited and fixed number of features for single-label classification in an online environment. Particularly, this algorithm deals with two diverse tasks of OFS: (i) OFS by training via complete inputs where a learner is permitted to access all the features to determine the subset of efficient features, and (ii) OFS by learning with partial inputs where the learner is permitted to access only a small number of features for each instance. The proposed technique attempts to solve each of the OFS problems and provides a theoretical analysis of the proposed OFS algorithms.

BenSaid et al. [205, 206] developed OFS method called multi-objective-based Automated Negotiation OFS (MOANOFS) based on the concept of automated negotiation using various learners to increase the classification performance of high-dimensional datasets. There are two levels of determination included in this method. It defines k learners that are the reliable ones from n learners, in the first level. In the second level, these chosen k learners are integrated with Multilateral Automated Negotiation to determine which is the best solution or which features are significant.

Lei et al. [207] initially, defined a neighborhood relationship to cope well with the issue of unequal distribution of data and introduced a weighted degree computational method taking into account the neighborhood structure of every object. Then developed a new OSFS method called OSFS-KW can select a small subset of relevant features and does not have to define any parameters in advance. Moreover, can handle the problem of learning from an imbalanced medical dataset.

Javidi et al. [199] considered the issue of SFS from the RS viewpoint and introduced a new approach called Streamwise feature selection based on RS (SFS-RS). This method applies the principle of feature relevance analysis to remove features that do not negatively affect the determination of the output feature and to manage the unspecified space of the feature. However, SFS-RS is a conventional approach based on the RS that cannot directly deal with numerical features. Bai et al. [208] developed a new method called Kernelized Fuzzy RS-based Online Hierarchical SFS (KFOHFS) For a Large Scale Hierarchical classification task. This method consists of three phases: (1) constructing a hierarchical structure data-oriented kernel-based fuzzy model with sibling technique, (2) selecting the significant online feature based on feature correlation analysis, and (3) removing the redundant online feature based on feature redundancy.

For multi-label data where instances may belong to more than one class several methods have been proposed for this scenario such as [210, 211]. Whereas Almusallam et al. [212] proposed unsupervised FS for streaming features (UFSSF) for effective dynamic selections of significant features. Further, Li et al. [213] developed a causality-based OSFS method with neighborhood conditional mutual information to address the problem of OFS. However, for high-dimensional imbalanced data, an OFS has been presented by Zhou et al. [214] to choose relevant features that can improve separability between the majority and minority classes.

3.2.2 Online group feature selection (OGFS)

The methods discussed in the previous subsection can successfully select a feature from the streaming feature only at the individual feature level without taking into account the group structures of online features. In the case of group structures, the selection of features at both the individual feature level and the group level is more preferred. In Group FS the selection of significant groups rather than individual features. Several OGFS methods are proposed in the literature to address the problem of FS at both the individual and group feature levels. This sub-section briefly describes the state-of-the-art existing OGFS methods and provides a summary of the most recent and efficient OGFS based objective and limitations in Table 8. Table 9 illustrates the comparative analysis that compares the OFS methods based on the category of FS (OIFS, OGFS), strategy, supervision, and classification type, datasets, the utilized classifiers for evaluating the performance, and the experiential platform.

Li et al. [215] developed a new online method called Group Feature Selection with Streaming Features (GFSSF) to select a feature from a given candidate feature subsets that generated dynamically and arrived so far at both the individual level and the group feature level in order to improve the classification performance. GFSSF is more efficient than existing algorithms that select only at the individual feature level. GFSSF algorithm consists of two levels of selection: feature and group level based on entropy and the information theory. At the feature level, the selection process is done from the same group and aims to select the most significant features from the features that arrived recently by exploring the relevancy between features and remove the redundancy. On the other hand, the group-level selection step aims to choose a group of feature sets that can cover as many as possible of the desired features' uncertainty at the minutest cost.

Wang et al. [216] proposed a novel online group approach to perform the selection process while keeping in mind the underlying group structure of online features called OGFS. Two steps are included in this algorithm: online inter-group selection and online intra-group selection. In the online intra-group selection step, aims to choose features in online environments by using a criterion based on spectral analysis that intends to select distinctive features in every group. After evaluating all feature sets, in the online inter-group step, OGFS intends to choose an appropriate subgroup based on the linear regression model then re-evaluates all the features chosen so far and eliminates the redundant features. Finally, this two-stage process keeps proceeding until no more features are arriving or some predefined stop conditions are satisfied.

Yu et al. [217] proposed an upgraded version of the SAOLA algorithm [194] to cope with features that reach by groups and then proposed a new technique called Group-SAOLA (GSAOLA) to handle the selection of features with group structure in an online situation. GSAOLA can pick a group of an online features that is scarce at the same time between the individual feature level and group feature levels. GSAOLA attempts to get a solution that is scattered at the different levels of both groups and individual level of features concurrently this would maximize its predictive performance for classification. FS for multi-label classification has got much attention in recent years. However, traditional multi-label FS are unable to deal with group structure of features and perform streaming feature selection concurrently. Therefore, to address this problem, Liu et al. [218] proposed a new algorithm to perform the FS process from multi-label FS keeping in mind the group structure of features in an online environment, this method is named Online Multi-label Group FS (OMGFS). It consists of two phases: Online Group Selection (OGS) and Inter Group Selection (IGS). In OGS phase, it selects the relevant group of features by using information theory measures. In IGS phase, the correlation between all the features that are selected reevaluate, then measure the interaction between features and perform redundancy analysis to eliminate the redundant features, finally come up with the final subset of optimal features. This two-phase process procedure continues till no more features arriving or stop conditions are satisfied. Zhou et al. [219] proposed OGFS, which considered the feature interaction within the

Table 8 Summary of OGFS methods

Technique	Objective	limitations	Ref.
GFSSF	To handle the selection of features from a given candidate feature subsets that generated dynamically and arrived so far at both the individual level and the group feature level in order to improve the classification performance.	Computational cost increases as data increases.	[215]
OGFS	To select online features considering the information of group structure of streaming feature as prior knowledge Two steps are included in this algorithm: an online intra-group selection and inter-group selection. They employed two measures for intra-group selection based on spectral analysis and presented a linear regression model to minimize the redundancy of features in inter-group selections.	In order to set parameters in advance, information is necessary, which is very difficult without prior knowledge.	[216]
Group-SAOLA	Upgraded the SAOLA algorithm [75] to cope with features that reach by groups in the online fashion and choose the most important features stream.	The optimum value of the significance threshold is difficult to achieve.	[217]
OMGFS	To perform FS process from a multi-label streaming feature keeping in mind the group structure of feature in an online environment.	Do not evaluate a redundant feature.	[218]

Table 9 Comparison of OFS methods

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
Grafting	[190]	**2003**	Support Vector Machine (SVM).	Used three DataSets (DS), (A and B are two synthetic datasets and C is a real-world problem taken from [221])	Matlab,	✓			✓	✓		✓	
Alpha investing	[191]	**2006**	- Applied stream-wise regression. - Tested using ten-fold cross-validation. - - SVM, NNET, (TREE).	Used six DSs taken from UCI repository [222], (Internet, Spect, Cleve, Wdbc, Ionosphere, and Wpbc) and Seven Bio-Medical DSts: (Aml, Ha Hung, Ctumor, Ocancer, Pcancer, Lcancer).	R	✓			✓	✓		✓	
OSFS	[193]	**2010**	K- Nearest Neighbour (K-NN), J48 and Random Forest(RF)	Used 10 public challenge DSs: Breast- Cancer, Hiva, Arcene, Dexter, Ovarian-Cancer, Anelon, Dorothea, Lymphoma, Nova, and Sido0.	C Language	✓			✓	✓		✓	
Fast-OSFS	[192]	**2013**	KNN AND J48	Used Ovarian-Cancer [223], Breast-Cancer [224], Lymphoma and Sido0 from [225], The Dexter, Arcene, Madelon, and Dorothea DSs are from the NIPS 2003 FS challenge [226].	C Language	✓			✓	✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
SAOLA	[194]	**2014**	J48 and KNN are available in the Spider Toolbox2 [227].	Used Ten high-dimensional DSs contain two biomedical DSs (Breast-Cancer and Hiva), Three NIPS 2003 DSs (Madelon, Dorothea, and Dexter), Two microarray DSs (leukemia and Lung-Cancer), Two text categorization data sets (Apcj-Etiology and Ohsumed), and the Thrombin DS from KDD Cup 2001. Four high-dimensional DSs collected from the Libsvm DS website: Url1, News20, Kdd2010, and Webspam.		✓			✓	✓		✓	
DIA-RED	[195]	**2013**	J48, Naive Bayes, and SVM	Used Six DSs: Dermatology, Backup-large, Mushroom, Tic-data2000, Splice, and Kr-vs-KP from UCI repository of machine learning DSs [222].	C#	✓		✓					

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
APSOFS	[196]	**2016**	Hyperpipe, Naviie Bayes, BayesNet, Decision Tree RF, SVM, and Neural Network	Used five DSs: Arcene, Dexter, Dorothea, Gisette, and Madelon from UCI [222].	MOA platform and java	✓		✓			✓	✓	
FRSA-IFS-HIS	[197]	**2015**	CART, Poly SVM and RBF SVM.	Used nine DSs include Wine, Heparits, Horse Iono, Credit, Credit (set), German, Sick, Bands. From the repository of machine learning DSs [222].	SQL Server 2008.	✓			✓				
MLFS	[198]	**2016**	ML-kNN	Used eight multi-label DSs include Business, Education, Emotions, Health, Reference, Science, Social, Yeast from [228].	Java			✓		✓		✓	✓

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
OS-NRRSAR-SA	[199]	**2016**	J48, kernel, Naive Bayes, SVM along with the RBF kernel function and JRip.	Used fourteen high-dimensional DSs include: Madelon, Dorothea, Dexter, and Arcene DSs are from [226]. The Cina0 and Sido0 DSs from the WCCI 2008 Causation and Prediction Challenges. The Arrhythmia and Multiple Features DSs are selected from the UCI [222]. Three synthetic DSs, Tm1, Tm3 and Tm2.	Not mentioned	✓			✓	✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
OSFSMI and OSFSMI-k	[200]	**2018**	DT, KNN and Naive Bayes	Used twenty-nine data sets from different domains include ALLAML, GLIOMA, Prostate_GE, SMK-CAN-187, ORL, GLA-BRA-180, OHSUMED-[illegible], Arrhythmia, MLL, Carcinom, Arcene, Madelon, orlraws10P, pixraw10P, warpAR10Pm, warpPIE10P, Yale, Extended, Yale, CINA0, Breastm, SRBCT, CNS, Lung cancer, Dexter, Lymphoma, Ovarian, ColonTumor, Nci9, Lung.	Matlab	✓			✓	✓		✓	
OFS-A3M	[201]	**2017**	KNN and SVM	Two data sets (WDBC, HILL VALLEY) from UCI. Seven microarray data sets (DLBCL, MLL, SRBCT, GLIOMA, PROSTATE, CAR, and LUNG2). (ARCENE) from NIPS 2003 DS.	Matlab.	✓			✓	✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
OFS-Density	[202]	**2019**	KNN, SVM, and CART.	Four data sets from UCI (WDBC, IONOSPHERE, HILL VALLEY, SONAR). Nine DNA microarray DSs (COLON, DLBCL, LYMPHOMA stdm, GLIOMA, PROSTATE-stdm, LUNG2, STRBCT, LEUKEMIA-std, MLL). (ARCENE) from NIPS 2003 DS.	Matlab	✓			✓	✓		✓	
OSFASW	[203]	**2019**	Decision Tree, KNN, SVM and Ensemble.	Used nine DSs from the NIPS 200: Madelon, Ionosphere, Dexter, Lung Cancer, Arcene, Dorothea, Prostate, and Leukemia. The lymphoma and Sido0 DSs. The Slyva, Lung, and Reged1 DSs from Causality Workbench. Reged1	Matlab	✓			✓	✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
OFS	[204]	**2014**	Online Gradient Descent (OGD) and KNN	Used six datasets from UCI: Magic04, Svmguide3, German, Splice, Spambase, and A8a. Two real text classification datasets (RCV1 and 20 Newsgroups).	Matlab	✓		✓		✓		✓	
MOANOFS	[205]	**2018**	SVM and KNN	Used nine datasets from UCI: A9a, Covtype, Gisette, Ijcnn1, Kddcup08, Magic, Mushrooms, Spambase, W8a, and Rcv1. Two real text classification datasets: 20News1 and 20newsgroup.	Not mentioned	✓		✓		✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
GFSSF	[215]	**2013**	K-NN, Naïve Bayes, C4.5, and Random Forest (RF).	Used five benchmark DSs from UCI: Ionosphere, Wdbc, Wpbc, Arrhythmia and Spectf. Five challenge high dimensional DSs: LUNG, CNS, DLBCL, ARCENE, and OVARIAN. Five UCI [222] DSs with group structures: NORTHIX, MULTI-FEATURES, ISOLET, MADELON, HILL-VALLEY.	Matlab	✓	✓		✓	✓		✓	
OGFS	[216]	**2015**	K-NN, J48 and RF in Spider Toolbox [227].	Used eight benchmark DSs from UCI: (Ionosphere, Wdbc, Spambase and Spectf). Microarray DSs (Colon, Prostate, Leukemia, and Lung cancer).	Not mentioned	✓			✓	✓		✓	

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
Group-SAOLA	[217]	**2016**	J48, SVM and KNN are provided in the Spider Toolbox [227].	Used ten high-dimensional DSs: Hiva, Ohsumed, Leukemia, Madelon, Breast-Cancer, Dexter, Apcj Etiology, Dorothea, Thrombin and Lung-Cancer. Four high-dimensional DSs: Webspam, Url1, News20and Kdd2010.	Matlab	✓	✓		✓	✓		✓	
OMGFS	[218]	**2018**	KNN and CART	Used eight multi-label datasets include: Business, Education, Emotions, Entertainment, Health, Reference, Arts, Science, Social, and Yeast from Mulan Library [228].	Not mentioned	✓	✓		✓	✓		✓	✓

Table 9 (continued)

Technique	Ref.	Year	Classifier	DataSet	Platform	Structure		Strategy		Supervision		Classification Type	
						Single	Group	Streaming Data	Streaming Feature	Supervised	Unsupervised	Binary	Multiclass
OSFS-KW	[207]	**2020**	KNN, SVM, and RF	Used a high-dimensional medical dataset: Hill Valley (Noise), Wdbc, Lymphoma, Dlbcl, Colon, Tumor, Car, Glioma, Srbct, Leu, Mll, Prostate, Lung Std, Arcene, Madelon, Lung, Breast Cancer, Ovarian Cancer, and Sido0.	Matlab	✓			✓	✓		✓	
SFS-RS	[199]	**2018**	The J48, kernel function, and kernel SVM with RBF.	Use four datasets from NIPS 2003: Dorothea, Arcane, Madelon, and Dexter. Five DSs from the UCI: Arrhythmia, Mf, Ionosphere, Wine, And Credit. Three synthetic DSs: Tm1, Tm2, and Tm3.	Not mentioned	✓			✓	✓		✓	
KFOHFS	[208]	**2020**	LSVM	AWAphog collected from Animals. Bridges from (UCI) library. Cifar image data sets. VOC and DD are protein datasets. F194 is a protein data set.	Not mentioned	✓				✓		✓	

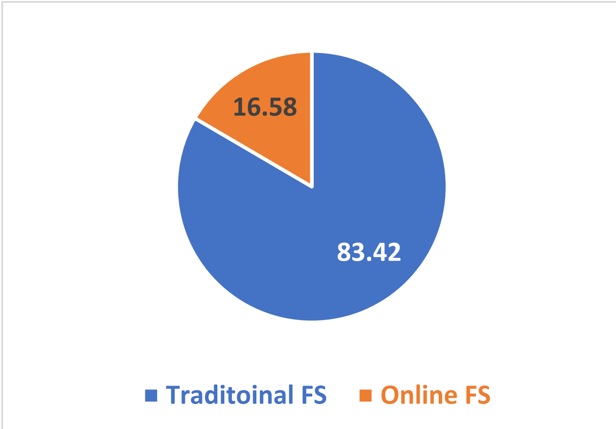


Fig. 3 Traditional and Online FS publications.

streaming groups. In [220] introduced OGFS, embedded within a new incremental technique, to choose optimal features from a feature stream.

4 Qualitative Analysis of the literature in FS

In this section, we discuss the quantitative analysis of the existing FS research papers publications based on the main category, sub-category, and timeline, in which we provide the answer to the following research question:

- How many FS research papers were recently published in each main category (Traditional and Online) and their sub-categories based on the timeline?

To answer the above research question, we have quantitatively analyzed the reviewed research papers in our taxonomy based on their main category, then we quantitative analyzed each main category based on the publication timeline and their sub-categories.

4.1 FS Publications based on main category (Traditional and Online)

The total number of reviewed research papers in our taxonomy is 187 papers, out of which 83.42% percent, (156) research papers are related to FS Traditional main category, whereas the rest 16.58% percent, 31 research papers are related to Online FS main category, this is clearly illustrated in Fig. 3.

4.2 Traditional FS publications

This sub-section quantitatively analyzes the publications of Traditional FS based on the publication timeline and subcategory:

4.2.1 Traditional FS publications based on Timeline

This taxonomy paper has reviewed 156 published research papers (83.42%) related to traditional FS, which have been published from 1994 to 2021. As illustrated in Fig. 4 we can observe that 21.15% of publications were published in the period between 1994 and 2012, which got the highest publication rate. Whereas from 2012 until 2017 the rate was between 2.56% and 6.41%, again, the number of publications has been increased in 2018 where the rate is 10.26%, but in 2019, the number of publications reduced

Fig. 4 Traditional FS publications based on Timeline

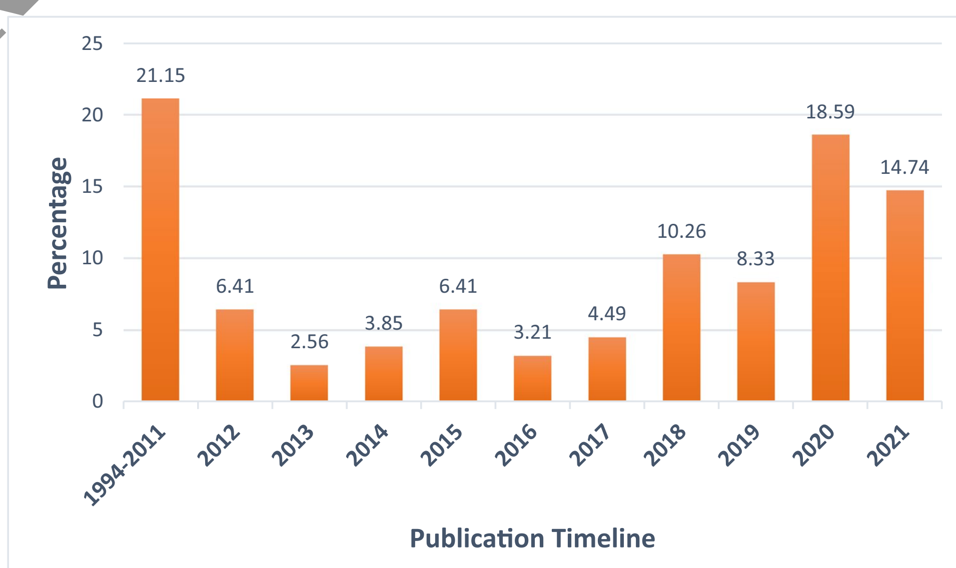
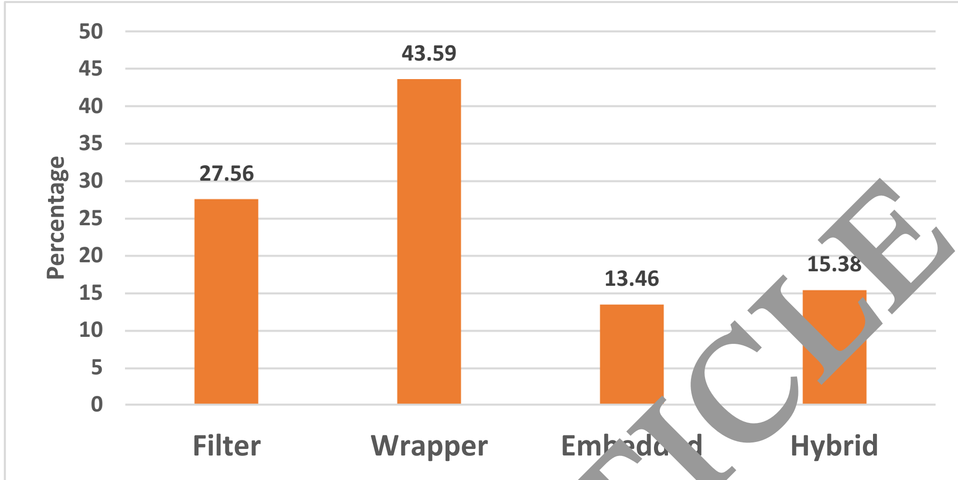


Fig. 5 Traditional FS Publications based on sub-Category



to 8.33%. Whereas in 2020 it goes higher to 18.59% but in 2021 it reduces to 14.74%. We can conclude that the higher number of publications were in 2020 and 2021.

4.2.2 Traditional FS Publications based on the sub-category

This taxonomy paper has classified the traditional FS method into sub-categories as follows: Filter, Wrapper, Embedded, and Hybrid. This section quantitatively analyzes the publications of traditional FS based on their sub-categories. As it is clearly observed from Fig. 5, the wrapper methods got 43.59 that is the highest publication rate. Whereas the filter has got 27.56%, which is the second rank, and the hybrid has got 15.38% and the Embedded has got 13.46% which is the lowest publication rate. We can conclude that the Embedded and hybrid got less attention from researchers.

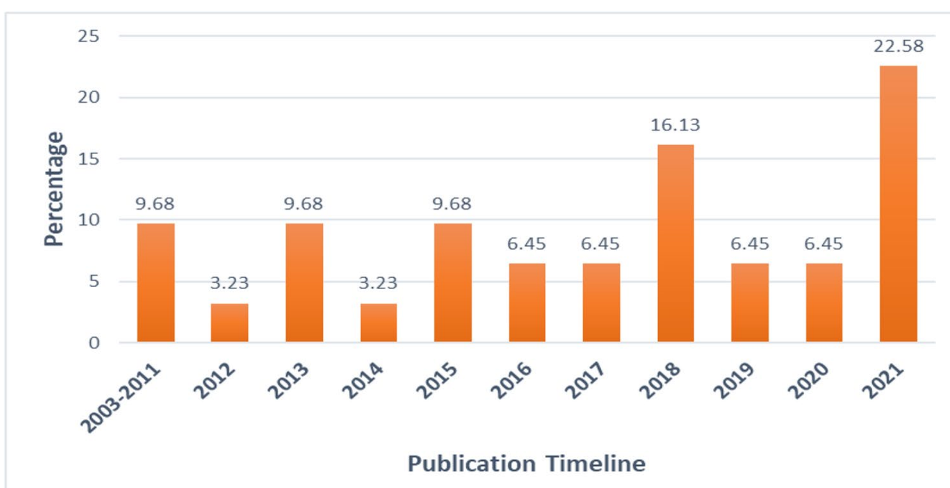
4.3 Online FS publications

This sub-section quantitatively analyzes the publications of online FS based on the publication timeline and sub-category:

4.3.1 Online FS publications based on Timeline

This taxonomy paper has reviewed 31 published research papers (16.58%) related to online FS, which have been published from 2003 to 2021. As illustrated in Fig. 6 we can observe that 9.68% present publications were published in the period between 2003 and 2012. Likewise, we can observe that 2013 and 2015 have got the same publication rate. In addition, 2012 and 2014 have got the same publication rate of 3.23% which is the lowest publication rate. Whereas the publication years such as 2016, 2017, 2019, and

Fig. 6 Online Publications based on Timeline.



2020 got the same rate of 6.45%. However, the number of publications has been increased in 2018 where the rate is 16.13%, and in 2021, the publication rate has increased to 22.58%, which is the highest rate. We can conclude that the highest rate of publications was in 2021.

4.3.2 Online publications based on Sub-Category

This taxonomy paper has classified the Online FS method into sub-categories as follows: Individual OFS and Group OFS. This section quantitatively analyzes the publications of online FS based on their sub-categories. As it is clearly observed from Fig. 7, the Individual OFS methods got 80.65, which is the highest publication rate. Whereas the Group OFS has got 19.35. We can conclude that the Group OFS has got less attention from researchers.

5 Experimental analysis

In this section, we conduct a comparative analysis of various FS methods, which experimentally analyze and evaluate their performance in terms of accuracy, precision, recall, F-measure, and the number of selected features. The experiments were conducted in such a way that, we initially compared four well-known methods from univariate and multivariate filter FS methods. The selected univariate method include Fisher score, and the selected multivariate methods are CFS, ReliefF, and mRMR. Then, some other six prominent wrapper methods were selected. These methods are as follows: PSO, GA, WOA, CS, HHO, and BA. Each method includes several adjustable parameters that need to be tuned before starting the feature selection process, all common and

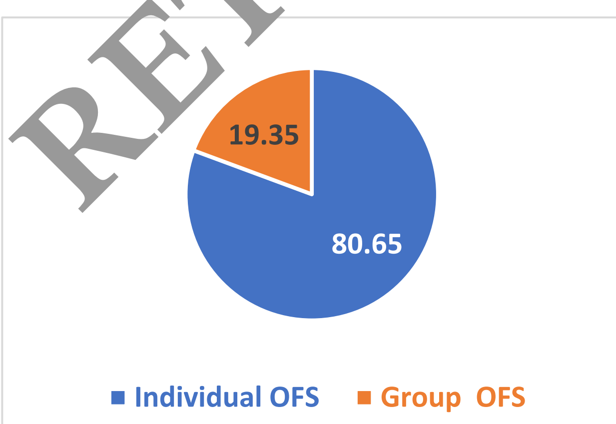


Fig. 7 Online Paper publications based on the sub-category.

Table 10 Parameters setup

Algorithm	Parameter settings	Value
Common parameters for all algorithms.	Population size	10
	Number of iterations	100
	Number of execution for each method	10
	k-value in *k*-nearest neighbor	5
	K for cross validation	5
PSO	Constants (C1, C2), Weights (W1).	C1 and C2 = [illegible], W1 = 0.9
GA	Crossover_Ratio (CR), Mutation_Ratio (MR)	CR = 0.8, MR = [illegible]
WOA	Constant (A).	A = 1
CS	Discovery_Rate (DR)	DR = 0.25
BA	Maximum_frequency (Fmax), Minimum_frequency (Fmin), Constants (α, β), Maximum_loudness (Lmax), Maximum_Pulse _Rate (MPR).	Fmax = 2, Fmin = 0, α = [illegible].9, β = 0.9, Lmax = 2, MPR = 1

specific setup parameters are illustrated in Table 10. To this end, four widely used classifiers including Support Vector Machine (SVM), Naïve Bayes (NB), K-Nearest Neighbor (KNN), and Random Forest (RF) were selected to evaluate the performance and efficiency of the foresaid FS methods.

The comparative study utilized 12 datasets; five of them were utilized for the comparison of filter methods, and the remaining were utilized for the comparison of a wrapper method. Where 70% of each dataset was utilized for training and 30% for testing, more details are given in next sub-section 5.1. All the experiments were carried out in the PyCharm 2020.2.3, on a system with configuration, Intel Core-i7 CPU with Windows 10 and 8 GB RAM.

This section includes three sub-sections, sub-section 5.1 provides a brief description of the experimental datasets, sub-section 5.2 briefly describes the evaluation metrics, finally, sub-section 5.3 analyzes and discusses the experimental obtained results.

5.1 Dataset description

This sub-section briefly describes the utilized dataset for the conducted experiments, 12 benchmark datasets with different properties were utilized in the experiments to show the effectiveness of the FS method. However, these datasets are from UCI Machine Learning Repository [222] and FS Repository of (ASU) Arizona State University [229]. The datasets are categorized into low dimensional datasets and high dimensional datasets. Moreover, each category can be of small sample data or large sample data. Table 11 presents a brief description of the datasets.

Table 11 Datasets summary

Data Set	#Features	#Instances	#Classes	Dimensionality	Sample size	Type (Data area)
Colon	2000	62	2	High	Small	Biological
GLI_85	22,283	85	2	Extreme High	Small	Biological
Leukemia	7070	72	2	High	Small	Biological
Lung	3312	203	5	High	Large	Biological
Prostate_GE	5966	102	2	High	Large	Biological
Arcene	10,000	200	2	High	Large	Mass Spectrometry
WaveformEW	40	5000	3	Low	Large	Physics
Breastcancer	9	699	2	Low	Large	Biological
BreastEW	30	569	2	Low	Large	Biological
Sonar	60	208	2	Low	Large	Biological
Arrhythmia	279	452	16	High	Large	Biological
Lung cancer	56	32	3	Low	Small	Biological

5.2 Evaluation Metrics

This sub-section briefly describes the evaluation metrics used to evaluate the efficiency of the selected FS methods for our experimental analysis. The evaluation was performed in terms of accuracy, recall, precision, F-measure, and the number of feature selections. For all experiments, each method was executed 10 independent runs, the averages of the obtained results were calculated, which is the average of results obtained from N runs for each method to all the metrics. All these metrics are described in the following equations:

Accuracy: the ratio of correctly predicted instances to the total number of observations. Accuracy is calculated using Eq. (1),

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{1}$$

Where the Average Classification Accuracy (ACC) is calculated as given in Eq. (2).

$$ACC = \frac{1}{N}\sum_{i=1}^{N} Accuracy_i \tag{2}$$

Precision: is the ratio of the correctly predicted positive observation to the total predicted positive observations. The precision is calculated using Eq. (3),

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Where Average of precision (AvgPrecision) is calculated as given in Eq. (4).

$$AvgPrecision = \frac{1}{N}\sum_{i=1}^{N} Precision_i \tag{4}$$

Recall: is the ratio of correctly predicted positive observations to all observations in the actual class. The recall is calculated using Eq. (5),

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

Where the average of recall (AvgRecall) is calculated as given in Eq. (6).

$$AvgRecall = \frac{1}{N}\sum_{i=1}^{N} Recall_i \tag{6}$$

F-measure: it is the weighted average of precision and recall, the F-measure is computed using Eq. (7),

$$F - Measure = \frac{2 * Precision * Recall}{Precision + Recall} \tag{7}$$

The Average of F-Measure (AvgFmeasure) is calculated as given in Eq. (8).

$$AvgFmeasure = \frac{1}{N}\sum_{i=1}^{N} F - Measure_i \tag{8}$$

The average number of selected features (A.NSF): this represents the average value of the selected features to the total number of features, which can be calculated as given in Eq. (9)

$$A.NSF = \frac{1}{N}\sum_{i=1}^{N} \frac{d_i}{D} \tag{9}$$

Where d_i denotes the number of selected features, D denotes the total number of features in the dataset and N is the number of executions.

5.3 Results and discussion

This sub-section reported the experimental results in terms of (accuracy, precision, recall, F-measures, and the number of selected features), the results discussion goes in this way, initially, we discuss the results of filter-based methods in sub-section 1, then, the results of wrapper based is discussed in sub-section 2.

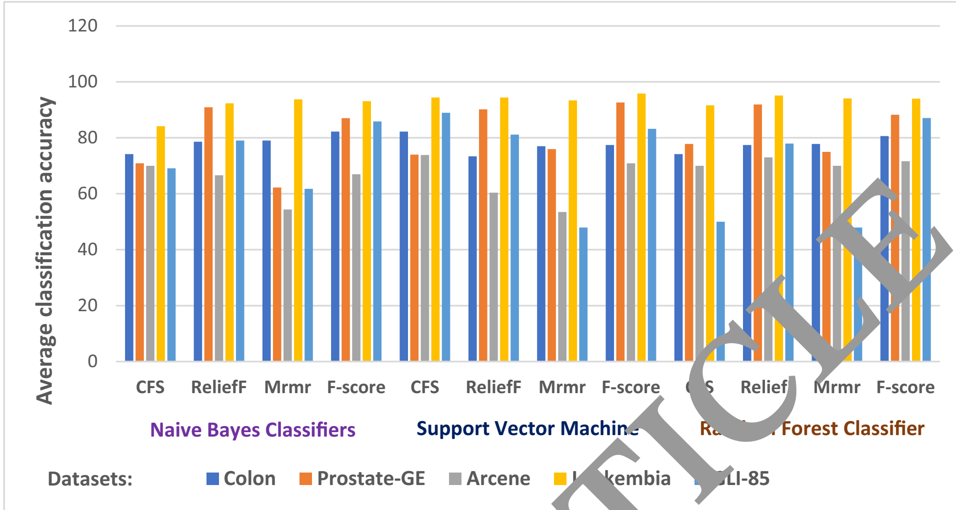
5.3.1 Filter based methods

In this sub-section, we evaluate the performance of chosen filter method (CFS, ReliefF, mRMR, Fisher score) for our experiments in terms of (accuracy, precision, recall, F-measure). Moreover, three classifiers were utilized (NB, SVM, RF) for each method, over five datasets (Colon, Prostate-GE, Arcene, Leukemia, GLI-85). We analyzed and discussed the obtained experimental results below:

Accuracy: To evaluate the performance of selected filter methods in terms of accuracy, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (2). The obtained experimental results are plotted in Fig. 8 which can be clearly observed; when the colon dataset is used the methods Fisher score and CFS got 82.25 over NB and SVM classifiers, which is the highest result. Whereas the ReliefF method has obtained 73.39% over SVM classifier which is the lowest result. However, in the case of the Prostate-GE dataset the method Fisher score got 92.65 with SVM classifier, which is the optimal score. Whereas the mRMR method has obtained 62.25 with the NB classifier which is the lowest score. Coming to the Arcene dataset the method ReliefF got 73.00 over the RF classifier, which is the highest result, whereas the mRMR method has obtained 53.05 over SVM classifier, which is the lowest result. When the Leukemia dataset is used the method Fisher score got 95.83 with SVM classifier which is the optimal result. Whereas the CFS method has obtained 84.16 over the NB classifier which is the lowest result. When the GLI-85 dataset has used the methods, CFS got 88.98 over the SVM classifier, which is the highest result. Whereas the mRMR method has obtained 47.95 over SVM and FR classifiers which is the lowest results. To this end, we can conclude that the optimal result is 95.83%, which was obtained from the Fisher score method with SVM classifier over the Leukemia dataset. Whereas the lowest result is 47.95 which was obtained from mRMR with SVM and RF over the GLI-85 dataset.

Precision: For evaluating the performance of selected filter methods in terms of precision, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (4). The obtained experimental results are plotted in Fig. 9 which can be clearly observed, when the colon dataset is used the method Fisher score got 82.90 over RF classifiers,

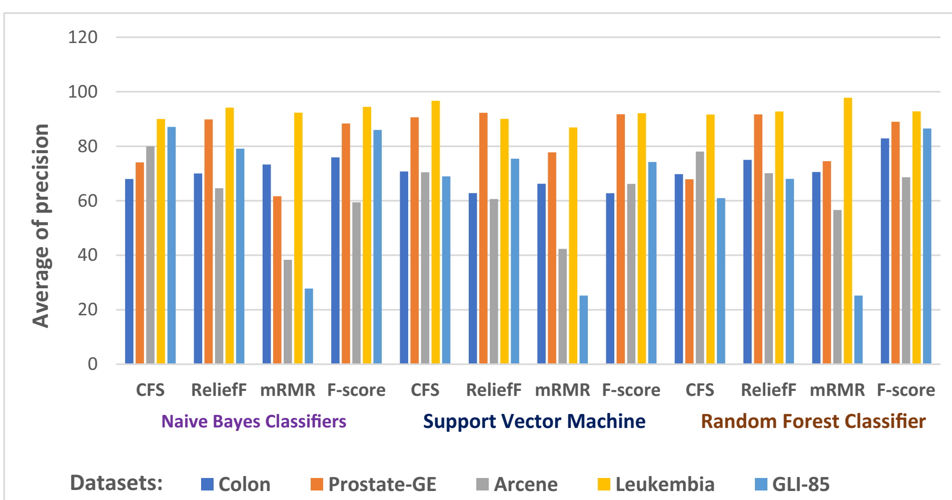
Fig. 8 CFS, ReliefF, mRMR, Fisher score Filter methods Average classification accuracy over different classifier and dataset



which is the highest results. Similarly, the same method has obtained results of 62.75% over the SVM classifier, which is the lowest result. When the Prostate-GE dataset is used the method Fisher score got 92.26 over the SVM classifier, which is the highest result. Whereas the mRMR method has obtained 61.65 over NB classifier which is the lowest results When the Arcene dataset is used the method CFS got 80.00 over NB classifier which is the highest results. Whereas the mRMR method has obtained 38.03 over NB classifier which is the lowest result. When the Leukemia dataset is used the method mRMR got 97.79 over the RF classifier which is the highest result. Whereas the mRMR method has obtained 86.92 over SVM classifier which is the lowest result. When the GLI-85 dataset has used methods CFS got 87.08 over NB classifier, which is the highest results, whereas the mRMR method has obtained 25.19 over RF and FR classifiers that are the lowest results. Finally, it can be concluded that the optimal result is 97.79% that was obtained from the mRMR method with RF classifier over the Leukemia dataset. Whereas the lowest result is 25.19 which was obtained from the same method but with different classifiers (SVM and RF) and GLI-85.

Recall: For evaluating the performance of selected filter methods in terms of recall, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (6). The obtained experimental results are plotted in Fig. 10 which can be clearly observed, when the colon dataset is used the methods Fisher score and Fisher score got 91.67 over SVM classifier, which is the highest results. Whereas the CFS method has obtained results of 65.83% over the RF classifier which is the lowest result. When the Prostate-GE dataset is used the method Fisher score got 93.38 over the SVM classifier, which is the highest result. Whereas the mRMR method has obtained 66.46 over the NB clas-

Fig. 9 CFS, ReliefF, mRMR, Fisher score Filter methods Average classification precision over different classifier and dataset



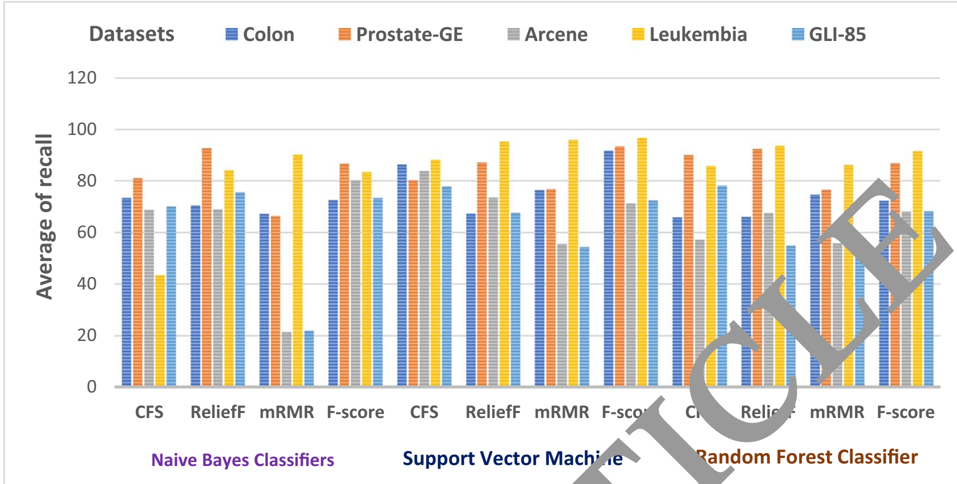


Fig. 10 CFS, ReliefF, mRMR, Fisher score Filter methods Average classification recall over different classifier and dataset

sifier which is the lowest result When the Arcene dataset is used the method CFS got 83.76 over SVM classifier which is the highest result. Whereas mRMR method has obtained 21.35 over NB classifier which is the lowest results. When the Leukemia dataset is used the method Fisher score got 96.67 over SVM classifier which is the highest result. Whereas CFS method has obtained 43.33 over NB classifier which is the lowest result. When the GLI-85 dataset is used the methods CFS got 78.09 over the RF classifier, which is the highest result. Whereas mRMR method has obtained 21.09 over NB classifier which is the lowest result. Finally, it can be concluded that the optimal result is 96.67% that was obtained from the Fisher score method with SVM classifier over the Leukemia dataset. Whereas the lowest result is 21.09 which was obtained from mRMR with NB classifier and over GLI-85.

F-measure: For evaluating the performance of selected filter methods in terms of F-measure, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (8). The obtained experimental results are plotted in Fig. 11 which can be clearly observed, when colon dataset is used the methods Fisher score and CFS got 77.10 over SVM classifier, which is the highest results, the same method has obtained results 62.39% over RF classifier that is the lowest results. When the Prostate-GE dataset is used the method Fisher score got 92.43 over SVM classifier, which is the highest result. Whereas the mRMR method has obtained 56.68 over NB classifier which is the lowest result When the Arcene dataset is used the method CFS got 79.65 over NB classifier which is the highest result. Whereas mRMR method has obtained 20.36 over NB classifier which is the lowest result. When the Leukemia dataset is used the method Fisher score got 94.29 over SVM classifier which is the highest result. Whereas CFS method has obtained 58.48 over NB classifier which is the lowest result. When the GLI-85 dataset is used the method Fisher score got 77.12 over NB classifier, which is the highest result. Whereas mRMR method

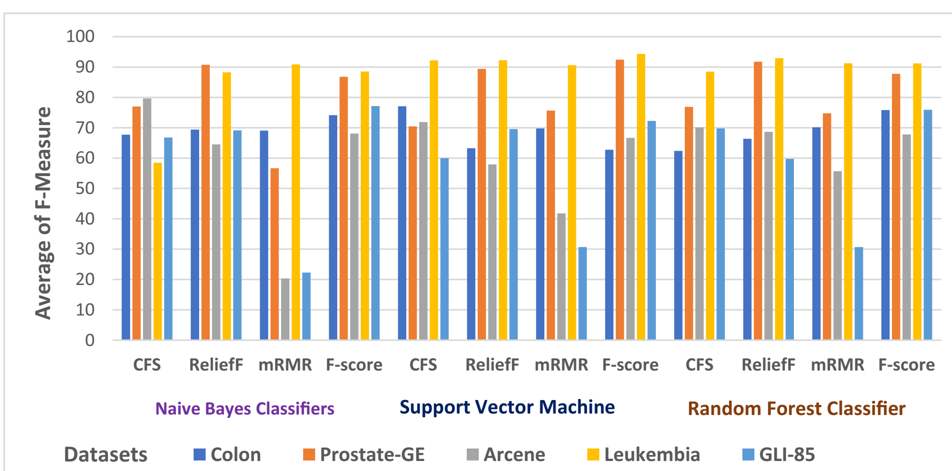


Fig. 11 CFS, ReliefF, mRMR, Fisher score Filter methods Average classification F-measure over different classifier and dataset

has obtained 22.28 over NB classifier which is the lowest result. Finally, it can bed concluded that the optimal result is 94.29% which was obtained from the Fisher score method with SVM classifier over the Leukemia dataset. Whereas the lowest result is 22.28 which was obtained from mRMR with NB and over GLI-85.

5.3.2 Wrapper based methods

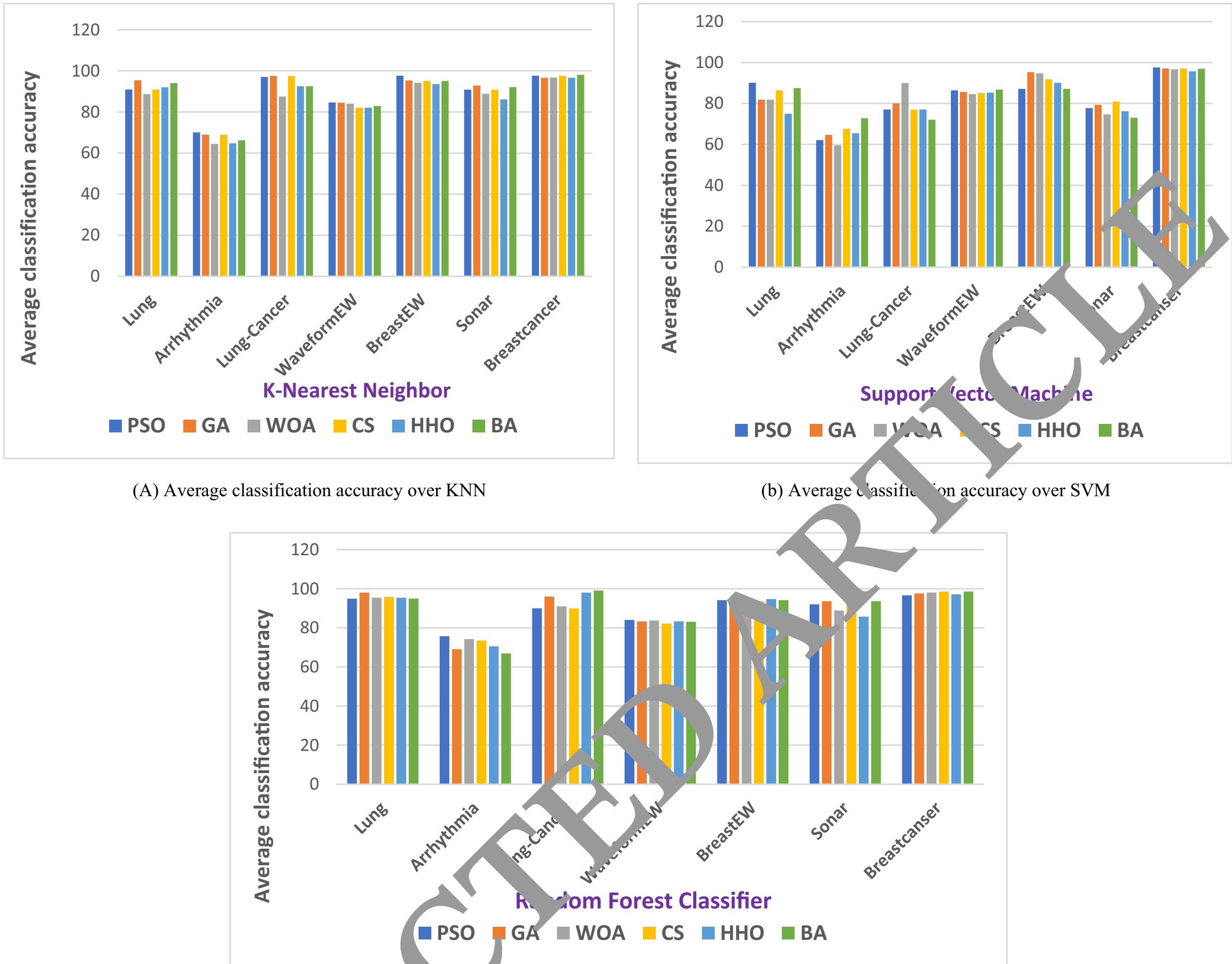
In this sub-section, we evaluate the performance of the selected wrapper methods (PSO, GA, WOA, CS, HHO, BA) for our experiments in terms of (accuracy and number of features). Three classifiers were utilized (KNN, SVM, RF) for each method, over five datasets (Lung, Arrhythmia, Lung-cancer, WaveformEW, BreastEW, Sonar, Breastcancer). We analyzed and discussed the obtained experimental results below:

Accuracy: For evaluating the performance of selected filter methods in terms of accuracy, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (2). The obtained experimental results are plotted in Fig. 12 (a), (b) and (c) which can be clearly observed when the KNN classifier is used, the method BA scored 98.09% over the Breastcancer dataset, which is the optimal result. Whereas the HHO method has obtained 64.71% over the Arrhythmia dataset, which is the lowest score as shown in Fig. 12(a). But the PSO, GA, CS, BA have scored 97.61%, 97.14%, 97.09%, 97.01% respectively when utilized with the SVM classifier over the Breastcancer dataset, and the WOA method has obtained 59.55% over the Arrhythmia dataset, which is the lowest results, as illustrated in Fig. 12(b). Coming to the RF classifier the methods BA and CS scored 99.08% and 98.57% over Lung-cancer and Breastcancer datasets respectively, which are the optimal results, whereas the method BA has obtained 66.91% over the Arrhythmia dataset which is the lowest results, as depicted in Fig. 12(c). Finally, it can be concluded that the optimal result is 99.08%, which was scored by the BA method with the RF classifier and over the Lung-cancer dataset. Whereas the lowest result is 59.55 which was obtained from WOA with SVM and over Arrhythmia.

Number of selected features: For evaluating the performance of selected wrapper methods in terms of the average number of selected features, we have run the experiment 10 times for each method over each classifier and dataset, then we calculated the average using Eq. (9). The obtained results are plotted in Fig. 13. (a), (b) and (c) over different classifiers (SVM, KNN, and RF) using various datasets. It can be clearly observed from Fig. 13, all methods were found to be good at selecting an average of small feature range between 3.02 to 37.32 when utilizing the (Lung-cancer, WaveformEW, BreastEW, Sonar, Breastcancer) with all classifiers. For Breastcancer all methods have got the best average results to range between 3.02 to 7.32. However, overall methods on the Sonar dataset the HHO was superior among all others using SVM which is capable of selecting only 10.12. Whereas, the WOA selects an average of 34.66 features when utilized with KNN. However, when using the WaveformEW dataset the CS was superior among all others using SVM, which select an average of 20.15 features. Whereas, the WOA select an average of 37.32 feature when utilized with RF. While using the Arrhythmia dataset all methods select a large number of features, among all methods the HHO was superior when used with SVM, whereas the GA method was superior with another classifier (KNN and RF) which obtained an average of 83.07 and 71.18 respectively. On the other hand, BA, CS, and PSO selected an average of 116.01 to 147.33 for all classifiers using the Lung dataset, which is the worst case. While other methods such as GA, WOA, and HHO select an optimal number of feature rage between 25.01 to 76.02 for all classifiers using the same dataset. Finally, it can be concluded that the best case is for GA over Breastcancer using the KNN classifier, whereas the worst-case goes to BA over the dataset Lung with KNN classifier.

The overall observation from the above experimental analysis, it can be observed that among the filter-based methods, the Fisher score was the more accurate method among others, when used with SVM and high dimensional dataset with small sample size such as Leukemia, whereas, when used with a high dimensional dataset with large sample size such as Arcene, the accuracy decreased. On the other hand, mRMR obtained the lowest accuracy among other methods, when used with extremely high dimensional datasets and small sample sizes such as GLI-85 and Arcene datasets with all classifiers. We can observe that whenever the dataset is high the accuracy gets decreased and vice versa. Generally, it can also be observed that all classifiers obtained good results for all the methods except mRMR that got low results over GLI-85 and Arcene in terms of accuracy, precision, recall, and F-measure.

In the wrapper-based methods, all methods with KNN and RF obtained good results for all the datasets except arrhythmia dataset accuracy is less for all methods, whereas for SVM all methods have scored good results range from 95.71 to 97.14 when used with Breastcancer, but the other



(A) Average classification accuracy over KNN

(b) Average classification accuracy over SVM

(c) Average classification accuracy over RF

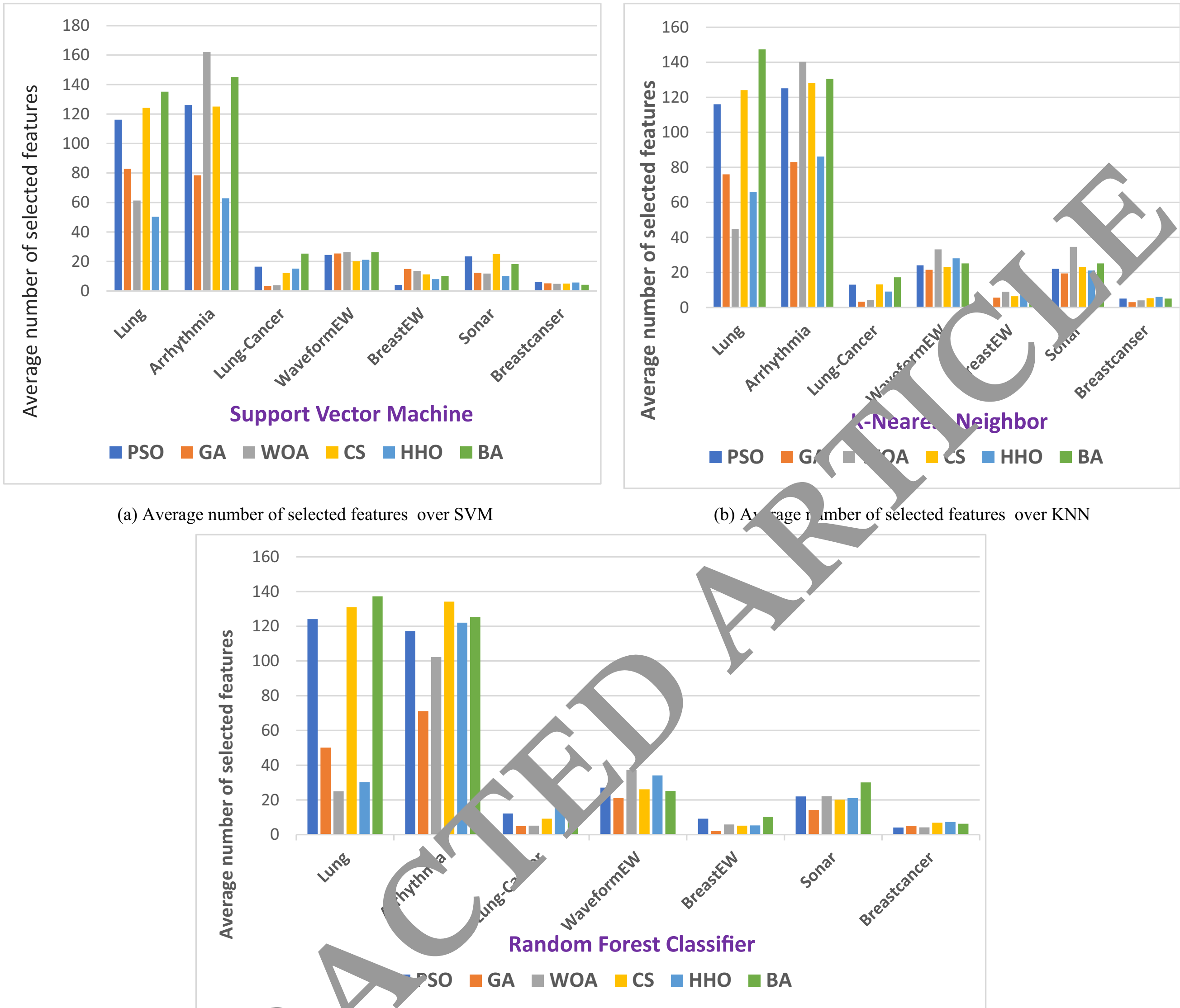
Fig. 12 Average classification accuracies of PSO, GA, WOA, CS, HHO, BA algorithms over different classifiers (SVM, KNN, and RF) using various datasets

datasets (Lung, WaveformEW, BreastEW,) obtained variant results range between 75% and 95.32, and (lung, Lung-cancer, Sonar) range from 72.5% to 90%. Finally, for the Arrhythmia, all the methods scored the least accuracy among other datasets, due to the high dimensionality and multi-class label dataset.

6 Research challenges and open issues

Although there have been a large number of research attempts to solve the FS problems, the unique characteristics of big data represent some challenges that have not been addressed yet, by the TFS task. Therefore, still, there are more works to be done in this area. In the next subsection, we will present challenges of FS for big data analytics that need more attention in future researches.

- ***Curse of dimensionality***: the curse of dimensionality is one of the significant challenges that researchers in the field often face when working on large volumes of data [230] and affects the efficiency of data processing in many data analytics and machine learning tasks. According to [231], the "curse of dimensionality" refers to the higher dimensions, where the number of samples needed to train the learning algorithm increases exponentially. Higher dimension data leads to the occurrence of noisy, irrelevant, and redundant data. Which causes overfitting of the prediction model and reduces the performance of the learning algorithm. To handle such problems, more research is needed on dimensionality reduction tech-



(a) Average number of selected features over SVM

(b) Average number of selected features over KNN

(c) Average number of selected features over RF.

Fig. 13 Average number of selected features of PSO, GA, WOA, CS, HHO, BA algorithms over different classifiers (SVM, KNN, and RF) using various datasets.

niques to reduce the dimension of big data during classification.

- ***Class imbalance:*** another significant problem is how to deal with imbalanced data. This issue occurs when the number of samples in some classes significantly be more than samples from the other classes [232]. The classes that have more samples than the other classes are called the majority or negative classes, while the other classes that have fewer samples are called the minority or positive classes. The minority classes are difficult to detect because they occur much less in frequency as compared to the other classes and they may carry important and useful knowledge. Which decreases the efficiency of classification models for machine learning? There is also a need for techniques for FS that can perform in the environment of imbalance data.
- ***Scalability:*** scalability is another important challenge that is taken into account when developing feature selection algorithms. It is defined as "the effect of increasing the size of a training set on the computational performance of an algorithm in terms of accuracy, training time, and allocated memory" [233]. With the exponential growth of data set size, scalability can be a major problem in today's FS that does not really scale well when dimensions are in the millions or more and degrade their efficiency gradually or even become ineffective in pro-

cessing. Scalable methods are then required, as current methods cannot handle a large number of features.

- ***Streaming feature selection:*** TFS approaches suppose that all features and data instances are available before the learning procedure takes place. However, another interesting scenario assumes that the features are produced dynamically and arriving one by one or group by group, and need to be processed when upon their arrival which is called SFS. This poses significant challenges to the TFS approach.
- ***Distributed processing:*** the main characteristic of data analytics is volume, which presents major challenges to machine learning because it takes time to learn from an enormous volume of data and discover a meaningful value, and the efficiency of current machine learning algorithms degrades as the volume of data increases. Big data comes in huge volumes, but due to privacy, legal data, and cost factors, it is difficult to move it to the central position. Parallel techniques of processing have been introduced to enhance the overall performance of big data analysis. Since FS methods will reduce the computing burden while implementing the data-mining model, it is interesting to explore a combination of the advantages of both feature selection and distributed processing.
- ***Feature selection for multi-label data:*** most of the existing FS algorithms mainly focus on the handling of single-label classification. However, in many scenarios the instances may have two or more labels. Thus, how to handle FS problem for multi-label data is a challenging issue.
- ***Unsupervised online feature selection:*** because of the unavailability of the class label, the unsupervised online FS methods are more challenging than the supervised online FS methods and achieving good prediction accuracy, which remains a challenge. As shown in Table 9 we observed that there is no more research in unsupervised online FS. Thus, there are still more research gaps that need to be addressed.

7 Conclusion and feature work

This article presented a comprehensive survey of recently proposed FS algorithms for both static and dynamic environments in various domains along with a taxonomy, which categorizes the FS method based on their search strategy, evaluation process, and feature structure. Initially, the study reviewed the existing traditional and online methods for FS; moreover, a qualitative analysis is performed to show the strengths and weaknesses of the existing traditional and online FS methods. The proposed taxonomy also quantitatively analyzed the existing FS techniques based on category, publication timeline. Further, an experiential analysis is conducted analyzing some of the most prominent FS methods from different categories with various classifiers on twelve benchmark datasets. Finally, based on the qualitative and experimental analysis, we highlighted our observations and some of the current challenges and open issues in FS.

This survey is meant to improve the efficiency in fast learning of the state-of-the-art FS methods, moreover, it will help researchers to understand and apply the vital characteristics of FS in Big Data. It also assists in finding out the limitations of current existing FS methods and potential research gaps. Definitely, this taxonomy will expand the researchers' minds and pave their way for possible new methods and impact future research. Through our taxonomy paper has provided a detailed and deep discussion on online methods, but still, the experimental analysis of this taxonomy is limited to filter and wrapper methods. In future work, we plan to carry out an experimental analysis that includes the online methods.

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